

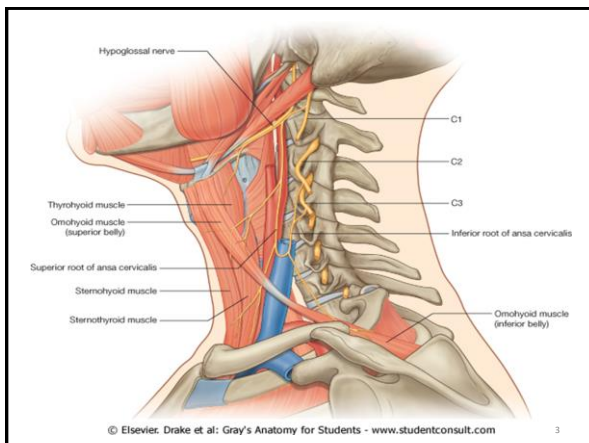
# Anatomy of The Neck

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## Introduction

- **The neck is located between**
  - ✓ the lower margin of the mandible above and
  - ✓ the **suprasternal notch and the upper border** of the clavicle below
- It is strengthened by the cervical part of the **vertebral column**
- Behind the vertebrae is a mass of **extensor muscles** and in front is a smaller group of **flexor muscles**
- Several vital structures, lack the bony protection, thus, **the neck is a well-known region of vulnerability**

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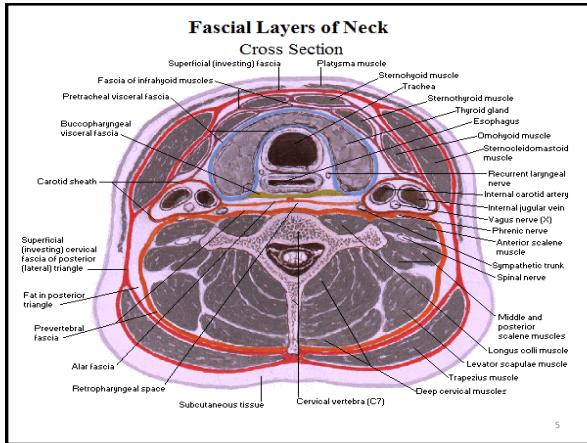


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## Introduction

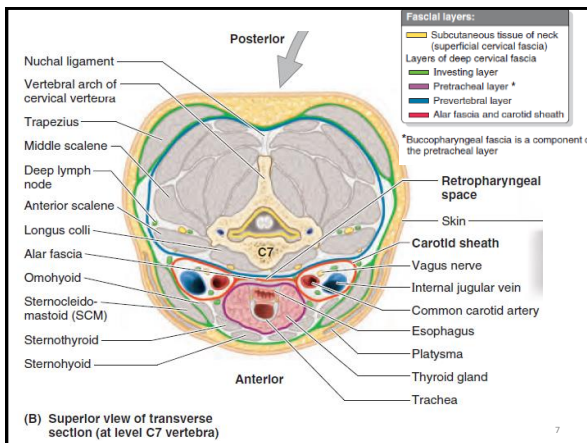
- It serves as **a major conduit** for structures passing through it in upward or downward directions
- **In the central region**
  - ✓ the larynx and the trachea, and
  - ✓ the pharynx and the esophagus
- **At the sides**
  - ✓ carotid arteries,
  - ✓ internal jugular veins,
  - ✓ the vagus nerve, and
  - ✓ the deep cervical lymph nodes

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## Fascia of the Neck

- Structures in the neck are surrounded by
  - ✓ a layer of subcutaneous tissue (**superficial fascia**) and
  - ✓ layers of **deep cervical fascia** involved in the compartmentalization of the cervical region
- The fascial planes determine the direction of infection spread



## Superficial cervical fascia

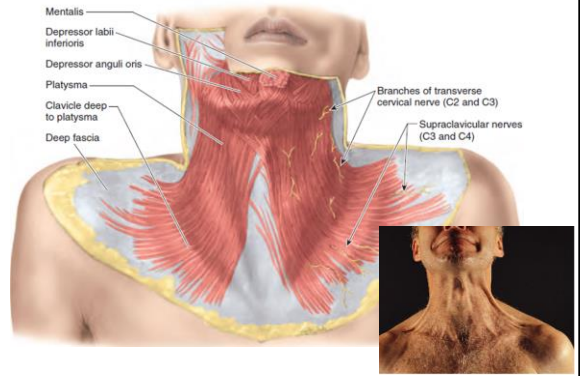
- It is also called cervical subcutaneous tissue
- It is
  - ✓ a layer of fatty connective tissue that lies under the dermis of the skin
  - ✓ usually **thinner than** in other regions, especially anteriorly.
- It contains
  - ✓ cutaneous nerves, blood and lymphatic vessels
  - ✓ superficial lymph nodes, and variable amounts of fat.
  - ✓ It also contains **platysma** (anterolaterally).

## The platysma muscle

- Thin, wide, located in the superficial fascia of the anterolateral aspect of the neck
  - ✓ which begins in the superficial fascia of the thorax, runs upwards to attach to the mandible & blend with the muscles on the face
  - ✓ Derived from 2<sup>nd</sup> pharyngeal arch
- **Origin** - skin and deep fascia over deltoid and pectoralis major
- **Insertion** – Inferior border of mandible, skin, and subcutaneous tissues of lower face
- **Nerve supply** – cervical branch of facial nerve
- **Action** – tenses skin of face e.g. during shaving

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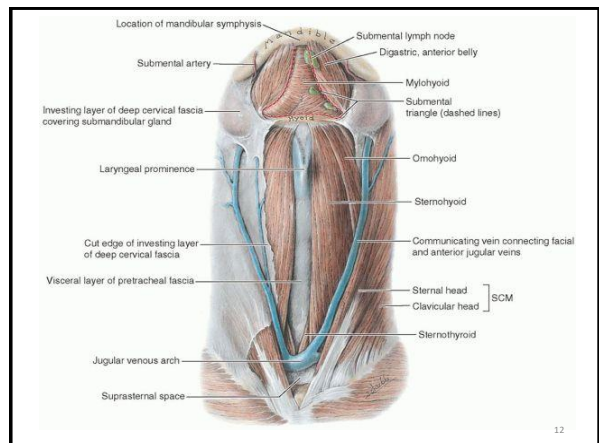
## The platysma muscle



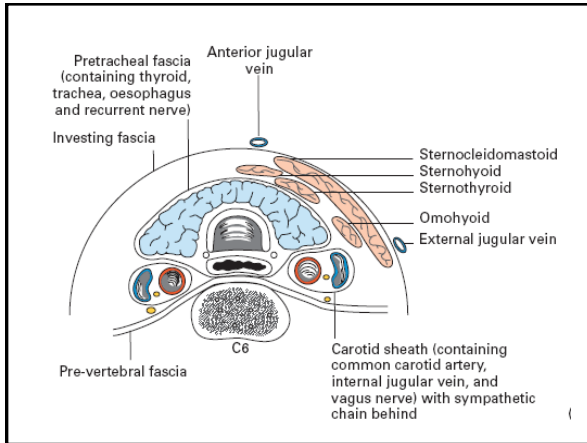
## Deep Cervical Fascia

- Deep fascia of the neck is arranged in three layers
- Surgically important, they make natural lines of cleavage through which tissues may be separated
  - ✓ give easy slippery movement of structures
  - ✓ Limit spread of pus

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## Deep Cervical Fascia

- The **deep cervical fascia** is organized into several distinct layers:

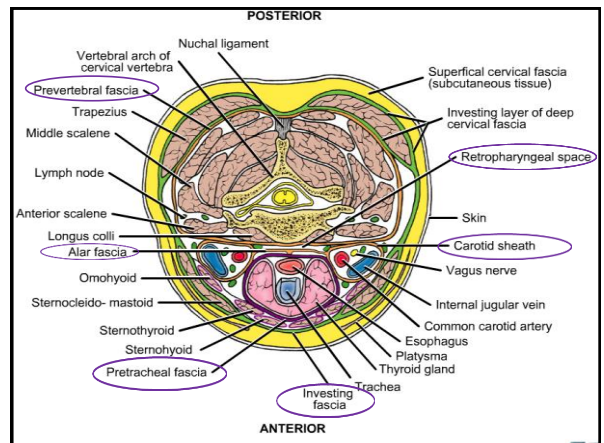
- ✓ **Investing layer**, which surrounds all structures in the neck;
- ✓ **Prevertebral layer**, which surrounds vertebral column & deep muscles associated with the neck/back;
- ✓ **Pretracheal layer**, which encloses the viscera of the neck (thyroid gland, trachea, esophagus);
- ✓ **Carotid sheaths**, which receive fascia from the other three fascial layers, surround the two major **neurovascular bundles** on either side of the neck.

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## Investing fascia

- Superficial layer of deep fascia
- **Location** – between superficial fascia and muscular layer; forms a collar around the neck
- **Forms the roof of anterior and posterior triangles of the neck**
- At the **four corners of the neck**, it splits into
  - ✓ **Superficial and deep layers** to enclose (invest) the **trapezius** and **sternocleidomastoid (SCM)** muscles

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## Investing fascia

### ▪ Attachments

#### – Superiorly

- superior nuchal line, mastoid process, zygomatic arch, inferior border of mandible, hyoid bone, spinous processes of cervical vertebrae
- Between angle of mandible and mastoid process, splits to **enclose parotid gland**
- Just inferior to its attachment to the mandible, it also splits to enclose the **submandibular gland**
- **The stylomandibular ligament** is a thickened modification of this layer

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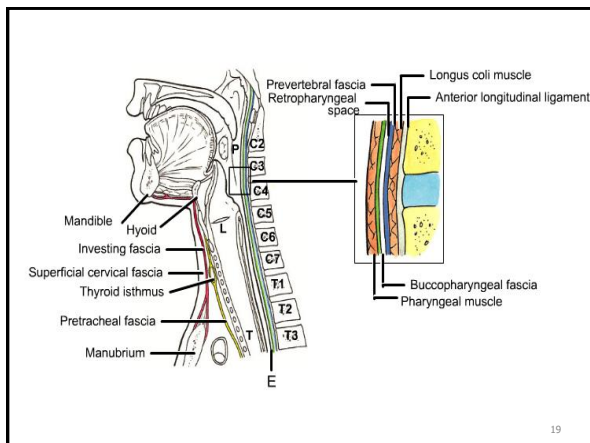
## Investing fascia

### ▪ Attachments..

#### – Inferiorly

- Manubrium of sternum, clavicle, acromion & spine of scapula
- Superior to sternum splits into two and attached to the anterior and posterior surfaces of manubrium
  - The interval between the two layers – **suprasternal space** contain;
    - ✓ sternal head of sternocleidomastoid
    - ✓ ends of anterior jugular veins
    - ✓ jugular venous arch
    - ✓ lymph nodes

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## Investing fascia

### ▪ Attachments..

#### – Anteriorly

- Symphysis menti and hyoid

#### – Posteriorly

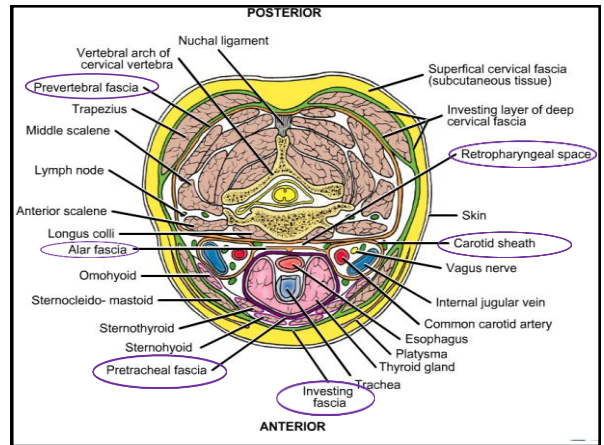
- Ligamentum nuchae and spine of C7

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## Pretracheal fascia

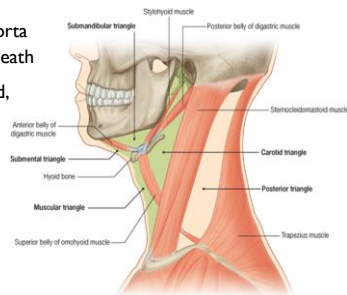
- Anterior to trachea, limited to the **anterior aspect** of the neck
- Extends from **hyoid bone** to the **thorax** to fuse with the **fibrous pericardium**
- Which includes:
  - A **thin muscular part**, which encloses the infrahyoid muscles, and
  - A **visceral part**, which encloses the thyroid gland, trachea, and esophagus and
    - ✓ is continuous posteriorly and superiorly with the **buccopharyngeal fascia of the pharynx**
    - ✓ laterally fuse with the **carotid sheath**

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## Pretracheal fascia

- Attachments**
  - Superior** – hyoid, thyroid cartilage and cricoid cartilage
  - Inferior** – arch of aorta
  - Lateral** – carotid sheath
- In the region of the hyoid, a thickening of the pretracheal fascia
  - ✓ forms a **pulley or trochlea** through which the intermediate tendon of the **digastric muscle** passes



## Prevertebral fascia

- Tubular sheath for the **prevertebral muscles** that surrounds the **vertebral column**
- Continuous with the **deep fascia forming the floor** of the posterior triangle of the neck
- Extends from** the base of the **skull** to the **3rd thoracic vertebra**
- Separated from pharynx by **retropharyngeal space**
- Superiorly**, it is fixed to the cranial base
- Inferiorly**, it blends with the **endothoracic fascia**
- Laterally**, it extends as the **axillary sheath**, which surrounds the axillary vessels and brachial plexus

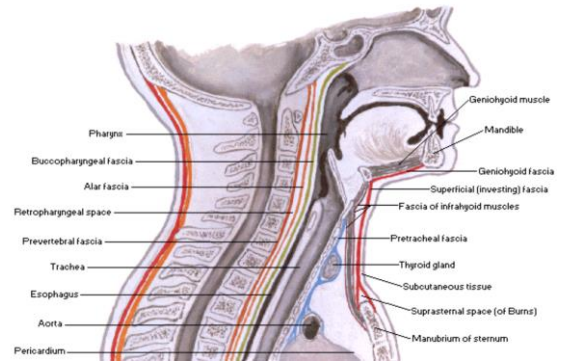
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## Applied anatomy

- The **investing layer** of deep cervical fascia helps prevent the spread of superficial skin abscesses into deeper parts of the neck
- If an infection occurs b/n the **investing layer** and the **muscular** part of the pretracheal fascia
  - ✓ the infection will usually not spread beyond the superior edge of the manubrium of the sternum.

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## Fascial Layers of Neck Sagittal Section



## Applied anatomy...

- Pus from the **posterior part** of the prevertebral fascia extends to the laterally to the posterior triangle and may pass to **axillary sheath**
  - ✓ The pus may perforate the prevertebral layer of deep cervical fascia and enter the **retropharyngeal space**
- Infections in the **retropharyngeal space** may also extend inferiorly into the **superior mediastinum**.
- Infection in the **head** may also **spread inferiorly**
  - ✓ posterior to esophagus into **posterior mediastinum**
  - ✓ Anterior to trachea to the **anterior mediastinum**

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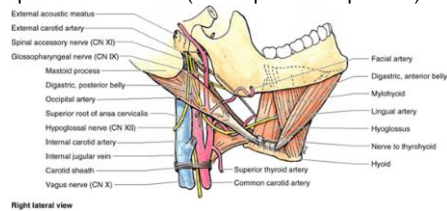
## The Carotid Sheath

- a tubular fascial investment
- It extends from the **cranial base to the root** of the neck
- This sheath blends with
  - ✓ **Anteriorly**, the **investing** and **pretracheal** layers of fascia and
  - ✓ **Posteriorly**, with the **prevertebral layer of fascia**

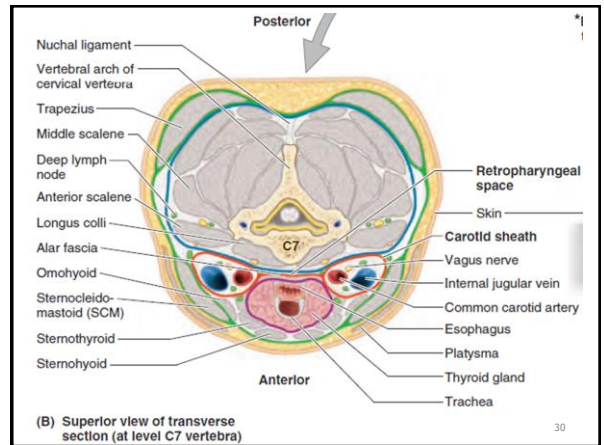
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## The carotid sheath contains

- ✓ The common and internal carotid arteries
- ✓ The internal jugular vein
- ✓ The vagus nerve (CN X)
- ✓ Some deep cervical lymph nodes
- ✓ The carotid sinus nerve, and
- ✓ Sympathetic nerve fibers (carotid periarterial plexuses)



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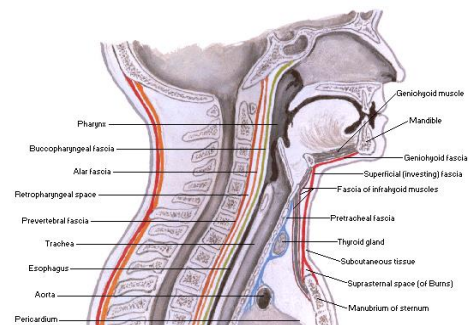
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## The Retropharyngeal Space

- Is the largest and most important **interfascial space** in the neck
- It is a **potential space** that consists of loose connective tissue
- **Located between**
  - ✓ the **visceral part of the prevertebral layer** and
  - ✓ the **buccopharyngeal fascia** surrounding the pharynx superficially
- The retropharyngeal space permits movement of the **pharynx**, **esophagus**, **larynx**, and **trachea** relative to the vertebral column during swallowing
- This space is **closed superiorly by the cranial base** and on **each side by the carotid sheath**
- It **opens inferiorly** into the **superior mediastinum**

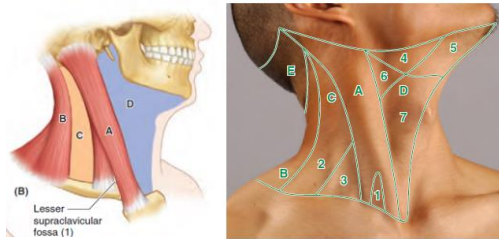
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## Fascial Layers of Neck Sagittal Section



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## The Cervical Regions



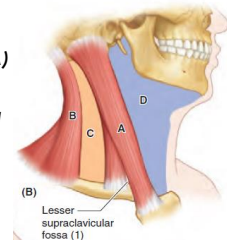
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## The Cervical Regions

- To allow clear communications regarding the location of structures, injuries, or pathologies, the neck is divided into regions

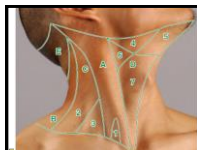
□ The four major regions are:

- (1). sternocleidomastoid region (A)
- (2). posterior cervical region (B)
- (3). lateral cervical region (C) and
- (4). anterior cervical region (D)



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## The Cervical Regions

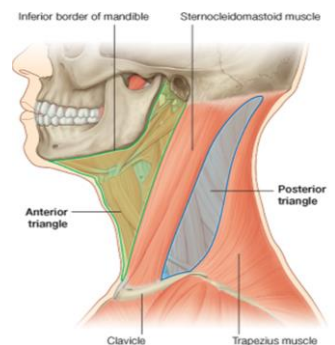


| Region <sup>a</sup>                              | Main Contents and Underlying Structures                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sternocleidomastoid region (A)                   | Sternocleidomastoid muscle; superior part of external jugular vein; greater auricular nerve; transverse cervical nerve                                                                                                                                                                                                                                                               |
| Lesser supraclavicular fossa (1)                 | Inferior part of internal jugular vein                                                                                                                                                                                                                                                                                                                                               |
| Posterior cervical region (B)                    | Trapezius muscle; cutaneous branches of posterior rami of cervical spinal nerves; suboccipital region or triangle (E) lies deep to superior part of this region                                                                                                                                                                                                                      |
| Lateral cervical region (posterior triangle) (C) |                                                                                                                                                                                                                                                                                                                                                                                      |
| Occipital triangle (2)                           | Part of external jugular vein; posterior branches of cervical plexus of nerves; spinal accessory nerve (CN XI); <sup>b</sup> trunks of brachial plexus; cervicodorsal trunk; cervical lymph node                                                                                                                                                                                     |
| Omo clavicular (subclavian) triangle (3)         | Subclavian artery (third part); part of subclavian vein (sometimes); suprascapular artery; supraclavicular lymph nodes                                                                                                                                                                                                                                                               |
| Anterior cervical region (anterior triangle) (D) |                                                                                                                                                                                                                                                                                                                                                                                      |
| Submandibular (digastric) triangle (4)           | Submandibular gland almost fills triangle; submandibular lymph nodes; hypoglossal nerve (CN XII); mylohyoid nerve; parts of facial artery and vein                                                                                                                                                                                                                                   |
| Submental triangle (5)                           | Submental lymph nodes and small veins that unite to form anterior jugular vein                                                                                                                                                                                                                                                                                                       |
| Carotid triangle (6)                             | Carotid sheath containing common carotid artery and its branches; internal jugular vein and its tributaries; vagus nerve; external carotid artery and some of its branches; hypoglossal nerve (CN XII) and superior root of ansa cervicalis; spinal accessory nerve (CN XI); <sup>b</sup> thyroid gland, larynx, and pharynx; deep cervical lymph nodes; branches of cervical plexus |
| Muscular (omotracheal) triangle (7)              | Sternohyoid and sternothyroid muscles; thyroid and parathyroid glands                                                                                                                                                                                                                                                                                                                |

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## Sternocleidomastoid Region

- Runs **superolaterally** from sternum and clavicle to mastoid process
- Landmark muscle** to divide the neck into anterior and posterior triangles



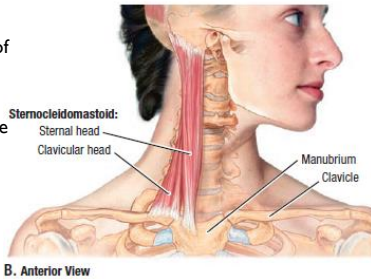
## Sternocleidomastoid Region

- **Origin** – two heads

- **Sternal head** – anterior surface of manubrium sternum

- **Clavicular head** – medial third of the upper surface of the clavicle

- Between the two heads is lesser **supraclavicular fossa**, overlying internal jugular vein



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## Sternocleidomastoid muscle

- **Insertion** – mastoid process & lateral half of superior nuchal line

- **Nerve supply** – spinal root of accessory nerve and branches from ventral ramus of C2 and C3

- **Action**

- one muscle acting alone - bend the head to the same side and rotate the face to the opposite side
  - the two muscles acting together– draw head forwards and flex the neck

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## Sternocleidomastoid muscle

- **Relations**

- **Superficial** – deep fascia, superficial fascia, platysma, external jugular vein, superficial cervical lymph nodes, greater auricular nerve, transverse cervical nerve, supraclavicular nerve
  - **Deep** – carotid sheath, muscles, vessels, nerves and deep cervical lymph nodes

- **Applied anatomy**

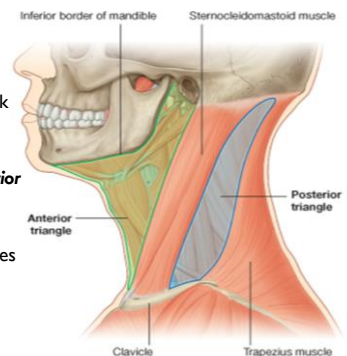
- **Torticollis** (wry neck) – caused by lesion of SCM muscle
  - ✓ It causes the head to tilt toward, and the face to turn away from, the affected side
  - ✓ like a normal unilateral SCM contraction,



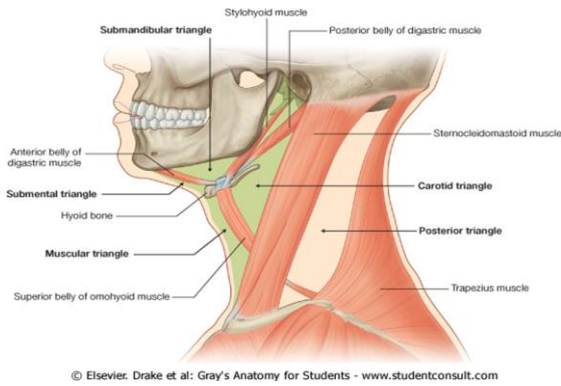
## Triangle of the Neck

- Each side of the neck is divided by **SCM** into

- ✓ anterior (**Anterior cervical region**) and
  - ✓ posterior triangles (**lateral cervical region**)



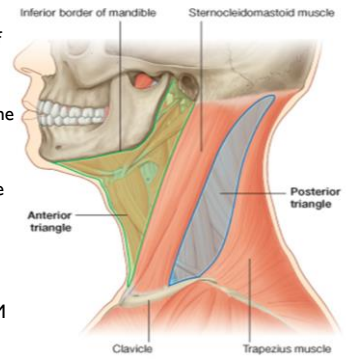
## Subdivision of Anterior Triangle



## Posterior Triangle of the Neck

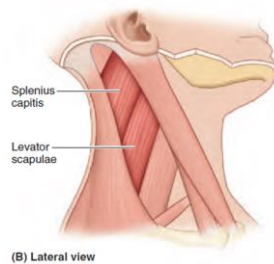
The posterior triangle of the neck is bounded:

- **Anteriorly** by the posterior border of the SCM
- **Posteriorly** by the anterior border of the trapezius
- **Inferiorly** by the middle third of the clavicle between the trapezius and the SCM



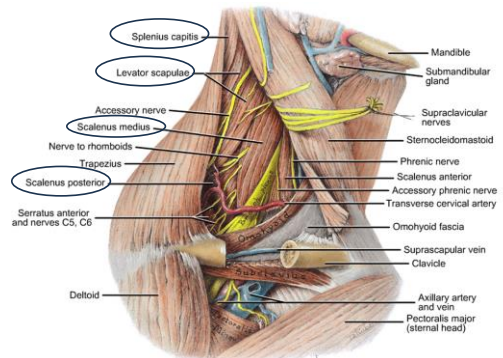
## Posterior Triangle of the Neck

- **An apex**, where the SCM and trapezius meet on the superior nuchal line of the occipital bone
- **A roof** - skin, superficial fascia & its contents, deep fascia
- **A floor** - prevertebral layer of deep fascia covering **scalenus posterior and medius, levator scapulae & splenius capitis**
  - ✓ some times inferior part of anterior scalene muscle



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## Posterior Triangle of the Neck

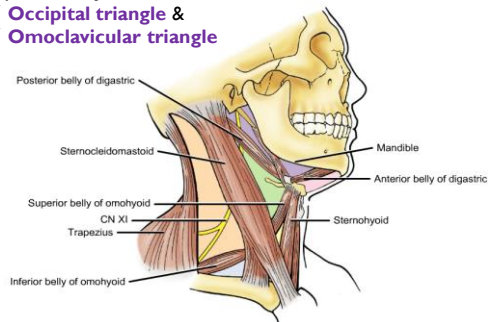


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## Posterior Triangle of the Neck

- **Subdivisions-** divided by inferior belly of **omohyoid** muscle into

- ✓ **Occipital triangle** &
- ✓ **Omoclavicular triangle**



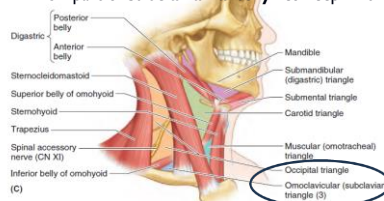
## Posterior Triangle of the Neck

### I. Occipital Triangle

- ✓ So called b/c **occipital** artery appear at its apex
- ✓ Spinal accessory nerve crosses it

### II. Omoclavicular (Subclavian) Triangle

- ✓ Indicated by supraclavicular fossa
- ✓ The inferior part of the **EJV** crosses this triangle superficially
- ✓ 3<sup>rd</sup> part of **subclavian artery** lies deep in it



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## Contents of Posterior Triangle

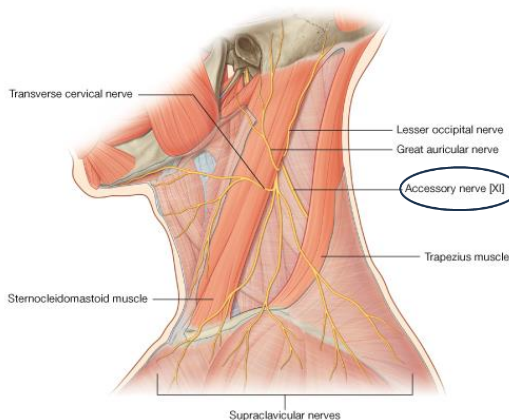
- **Muscles-** inferior belly of omohyoid

### ■ Nerves:

- Spinal root of **accessory nerve: supply SCM & trapezius**
  - enter in the middle of the triangle from the posterior border of the SCM
  - divide posterior triangle into equal parts
  - descend parallel to levator scapulae
  - exit above the clavicle by passing deep to the trapezius muscle

### II. Brachial plexus

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## Contents of Posterior Triangle

### III. Cervical plexus

- ✓ Formed by the ventral rami of C1-4

#### ▪ Branches

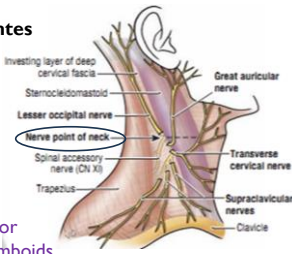
##### ✦ Gray rami communicantes

##### ✦ Cutaneous branches

- ✦ lesser occipital,
- ✦ greater auricular,
- ✦ transverse cervical and
- ✦ supraclavicular

##### ✦ Muscular branches –

- ✦ Phrenic, nerves to levator scapulae, trapezius, rhomboids, serratus anterior, and subclavius



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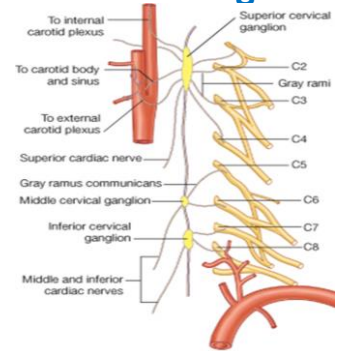
## Contents of Posterior Triangle

### A. Communicating

- ✓ **Gray rami:-** from superior cervical sympathetic ganglia to C1-4 nerves

- ✓ **Branch from C1** to the hypoglossal, supply geniohyoid & Thyrohyoid

- It leaves the hypoglossal nerve to form **superior root** of **ansa cervicalis**



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## Contents of Posterior Triangle

### B. Cutaneous

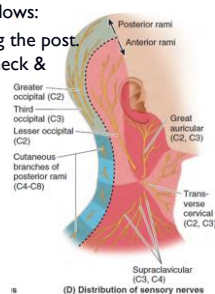
- ✓ Enter the triangle from the middle of posterior border of the triangle, pierce the deep fascia
- ✓ Cutaneous division further divided as follows:

- I. Lesser occipital:-** passes upward along the posterior border of the SCM, supply skin of the neck & lateral part of the back of the neck

- II. Great auricular:-**

- ascends up & forward behind and parallel to EJV & supply

- ✓ Skin of the neck
- ✓ Skin of posteroinferior parts of the face
- ✓ Skin of the mastoid process
- ✓ Parts of auricle



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(D) Distribution of sensory nerves

## Contents of Posterior Triangle

### III. Transverse cervical n:

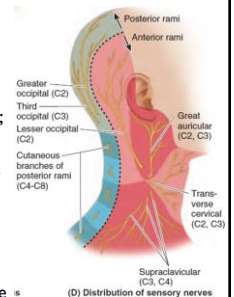
- ✓ Passes forward superficial to SCM muscle, divide into upper and lower fibers

### IV. Supraclavicular nerve:

- ✓ **arise as a single trunk & divide** into medial, intermediate and lateral; passes downward and

- ✓ pierce the fascia above the clavicle & **supplies**

- Skin of the neck
- Intermediate fibers:- chest up to 2<sup>nd</sup> rib in the mid line
- Lateral fibers:- skin of the shoulder girdle & over the upper 1/2 of the deltoid



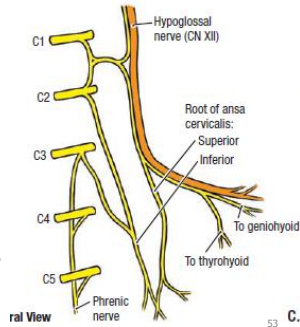
(D) Distribution of sensory nerves

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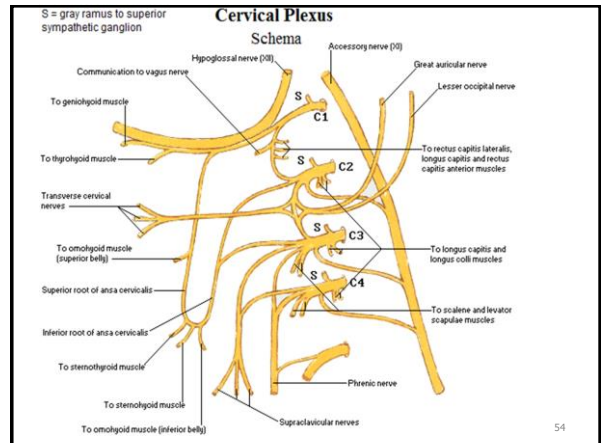
## Contents of Posterior Triangle

### C. Muscular

- ✓ **C2** to sternocleidomastoid muscle
- ✓ **C2 & 3** for **inferior root** of the **ansa cervicalis** & unite with the superior root;
  - supply infrahyoid ms. **except thyrohyoid m.**
- ✓ **C4&5** to levator scapulae
- ✓ **C3,4&5** phrenic nerve to diaphragm



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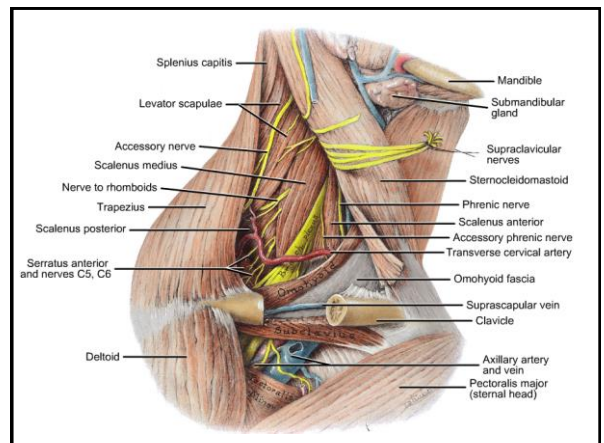
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## Contents of Posterior Triangle

### III. Brachial plexus

- ✓ The **roots and trunks** are found in the lower part of the posterior triangle
- ✓ **Divisions** are behind the clavicle while the **cords** are in the axilla
- ✓ Formed from **C5-8 & T1** spinal nerve roots
- **Muscular Branches in the neck**
  - ✓ **C5**-phrenic & dorsal scapular nerve
  - ✓ **C5&6**-nerve to subclavius and suprascapular
  - ✓ **C5-7**-long thoracic nerve

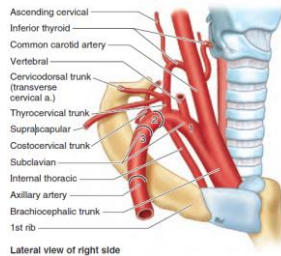
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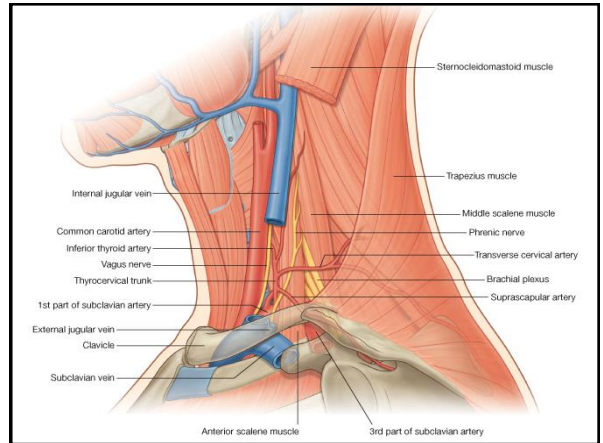
## Contents of Posterior Triangle

### Arteries

- ✦ **Transverse cervical artery – (cervicodorsal trunk)** inferior part of the triangle
- ✦ **Suprascapular artery** - inferior part of the triangle
- ✦ **Occipital artery** – cross apex of the triangle
- ✦ **Third part of the subclavian artery** – passes behind scalenus anterior over first rib
  - Can be compressed against the rib to control bleeding



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## Contents of Posterior Triangle

### Veins

- ✦ Subclavian
- ✦ Terminal part of external jugular vein
- ✦ Terminal part of anterior jugular vein
- ✦ Transverse cervical
- ✦ Suprascapular

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## Vessels in the posterior triangle

### Third part of the subclavian artery

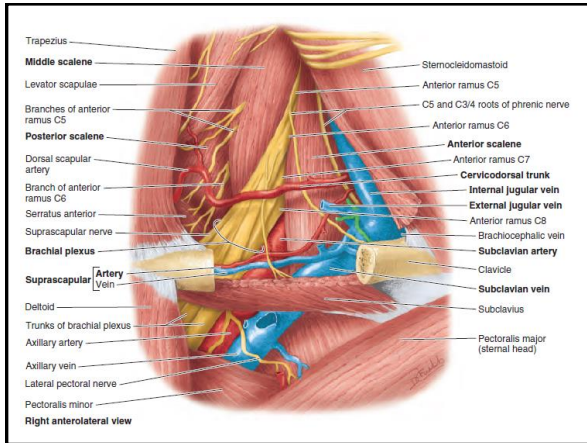
- Begins in the **lateral border of the scalenus anterior** one finger breadth above the clavicle
- Ends at the **outer border of the first rib** (opposite the middle of the lower border of clavicle) to become axillary artery

#### Relations

##### – Anteriorly

- **In the posterior triangle**- Terminal part of external jugular vein and nerve to subclavius
- **Behind the clavicle** – subclavius muscle, suprascapular artery & subclavian vein

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## Vessels in the posterior triangle

### Relations...

#### – Posteriorly

- lower trunk of brachial plexus
- Scalenus medius muscle

#### – Inferiorly

- Medially – apex of the lung and cervical pleura
- Laterally - first rib

#### – Above and laterally

- Upper and middle trunk of the brachial plexus
- Inferior belly of the omohyoid muscle

**Branches** – occasionally dorsal/descending scapular artery; substitute for deep branch of transverse cervical artery

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## Vessels in the posterior triangle

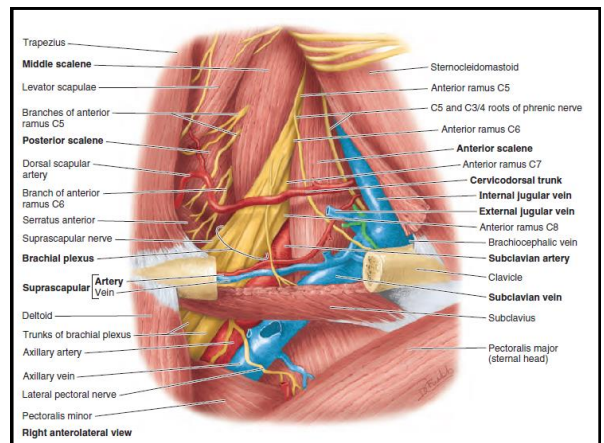
### ▪ Transverse cervical artery

- Branch from the **thyrocervical trunk**
- **Course** – deep to the SCM muscle to enter **posterior triangle**; continues laterally in front of scalenus anterior and trunks of brachial plexus

#### – Branches

- **Superficial** – deep to trapezius
- **Deep** – deep to levator scapulae and rhomboids along the medial border of the scapula; **anastomoses** with **subscapular** artery at the inferior angle of the scapula

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## Vessels in the posterior triangle

### ▪ Suprascapular artery

- A branch of **thyrocervical trunk**
- **Course**
  - ✓ run downward and laterally deep to SCM muscle
  - ✓ descends behind the middle third of the clavicle in front of brachial plexus & 3rd part of the subclavian artery up to the scapula
  - ✓ passes **above the suprascapular ligament** to reach supraspinous fossa
  - ✓ passes through the **spinoglenoid notch** to enter infraspinous fossa where it **anastomoses with circumflex scapular artery**

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## Vessels in the posterior triangle

### ▪ Subclavian vein (it is not really a content)

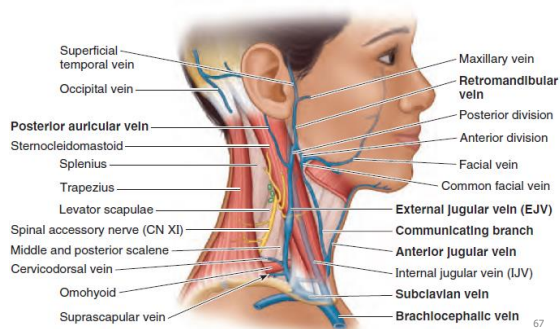
- **Begins** - at the outer border of 1st rib
- **Ends** – medial margin of scalenus anterior joining **internal jugular vein** to form innominate vein and accompany 2nd and 3rd part of subclavian artery below the artery

### ▪ External jugular vein

- **Begins** – below and behind the **angle of mandible** by union of
  - ✓ posterior auricular vein and
  - ✓ posterior division of retromandibular vein

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## Vessels in the posterior triangle



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## Vessels in the posterior triangle

### ▪ External jugular vein...

- **Course**
  - in the superficial fascia superficial to the SCM
  - **Pierce the deep fascia** of the roof of the posterior triangle of the neck above the clavicle behind the SCM muscle to enter posterior triangle
  - In the posterior triangle it is superficial to the brachial plexus and 3rd part of the subclavian artery
- **End** – behind the clavicle **joining subclavian vein**
- **Tributaries**
  - ✓ Anterior jugular
  - ✓ Transverse cervical (cervicodorsal)
  - ✓ Suprascapular veins

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## Muscles forming floor of posterior triangle

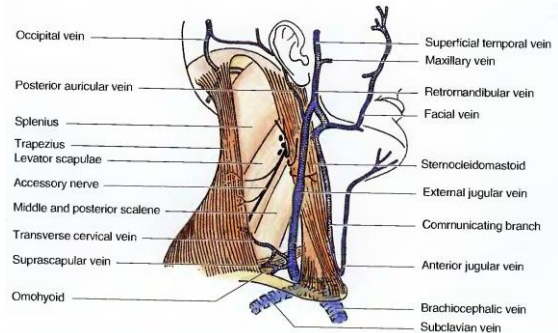
### Form floor of the posterior triangle

#### Splenius capitis

- Origin – lower half of ligamentum nuchae, spines of C7, T1-6
- Insertion – lateral third of superior nuchal line and mastoid process
- Innervation – dorsal rami of middle cervical spinal nerves
- Action
  - one muscle – bend the head and neck to the same side & turns the face to the same side
  - two muscles - extend the head and neck

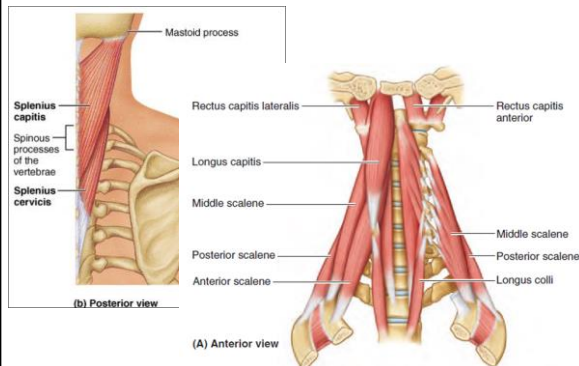
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## Muscles forming floor of posterior triangle



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## Muscles forming floor of posterior triangle



## Muscles forming floor of posterior triangle

### Levator scapulae

- Origin – transverse process of C1 to C4
- Insertion – superior part of medial border of scapula
- Innervation – dorsal scapular and cervical spinal nerves
- Action – elevates scapula

### Scalenus medius

- Origin - transverse process of C2 to C7
- Insertion – superior surface of first rib
- Innervation – ventral rami of cervical spinal nerves (C3-C8)
- Action – flex neck laterally and elevates first rib

### Scalenus posterior

- Origin - transverse process of C4 to C8
- Insertion – external border of second rib
- Innervation – ventral rami of cervical spinal nerves (C3-C8)
- Action – flex neck laterally and elevates second rib

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### Muscles forming floor of posterior triangle

| Muscle             | Origin                                    | Insertion                                           | Innervation                              | Action                                                           |
|--------------------|-------------------------------------------|-----------------------------------------------------|------------------------------------------|------------------------------------------------------------------|
| Splenius capitus   | Inferior 1/2 ligamentum N, spinous p.T1-6 | Mastoid P. superior nuchal line                     | Dorsal rami middle cervical spinal N     | Lateral flex & rotate head to the same side<br>Extend- bilateral |
| Levator scapulae   | Post T of Transverse P C1-4               | Superior part, medial border of scapula             | Dorsal scapular & cervical spinal nerves | Elevate scapula                                                  |
| Scalenus Posterior | Transverse P C4-8                         | 2 <sup>nd</sup> rib external part                   | Ventral R C3-8                           | Lateral flexion, elevate 2 <sup>nd</sup> rib                     |
| Scalenus Medius    | Transverse P C2-7                         | 1 <sup>st</sup> rib superior surf subclavian groove | Ventral R C3-8                           | Lateral flex, elevate 1 <sup>st</sup> rib                        |

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### Muscles forming floor of posterior triangle

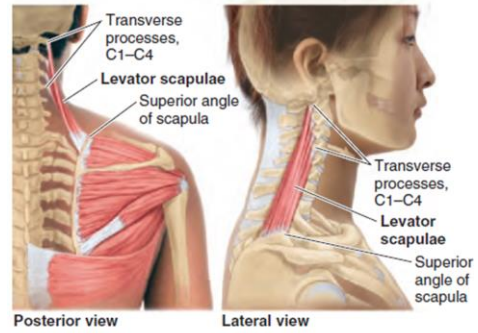


FIGURE 6.27. Levator scapulae.

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### Superficial cervical lymph nodes

- Along external jugular vein
- receive afferent from parotid, occipital, mastoid lymph nodes and from muscles and skin
- efferent pass to supraclavicular and deep cervical lymph nodes

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### Anterior Triangle of the Neck (Anterior Cervical Region)

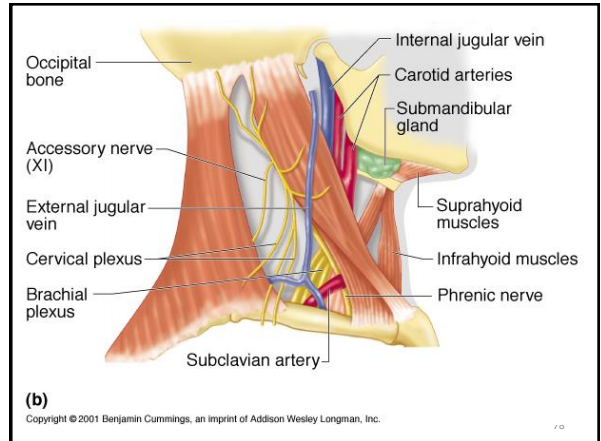
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## Anterior triangle of the neck

### ■ Boundaries

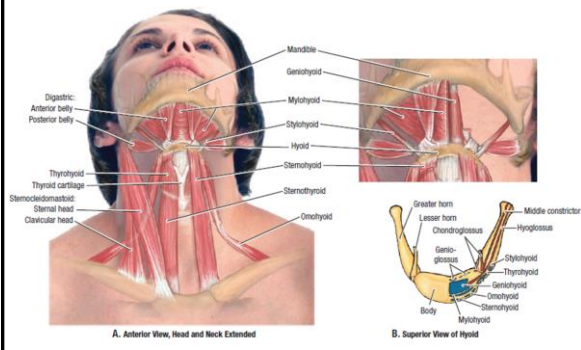
- **Anterior** - median line of the neck from chin to manubrium
- **Posterior** - anterior border of the sternocleidomastoid
- **Base (superior)** - inferior border of mandible and plane connecting angle of the mandible to mastoid process
- **Apex** - at jugular notch
- **Floor** - formed by pharynx, larynx and thyroid gland

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## Anterior triangle; Suprahyoid and infrahyoid muscles



## Contents

### ■ Hyoid muscles: 8 (Suprahyoid and infrahyoid muscles)

#### Suprahyoid muscles = 4

#### • Mylohyoid muscle

- Forms the floor of the mouth below tongue
- Origin - mylohyoid line of the mandible
- Insertion - body of hyoid
- Nerve supply - mylohyoid nerve (V3)
- Action - elevate the hyoid bone

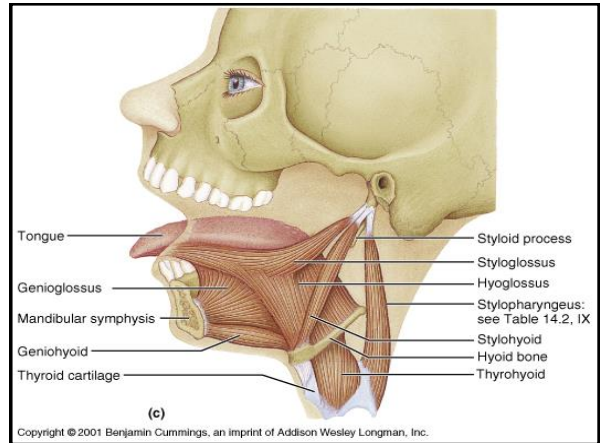
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## Contents

### ▪ Geniohyoid muscle

- Superior to mylohyoid; reinforce floor of mouth
- **Origin** – from inferior genial tubercle of mandible
- **Insertion** – into the anterior surface of the body of the hyoid bone
- **Nerve supply** – C1 through the hypoglossal nerve
- **Action** – pull the hyoid bone up and forward, shorten mouth floor and widens pharynx

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## Contents

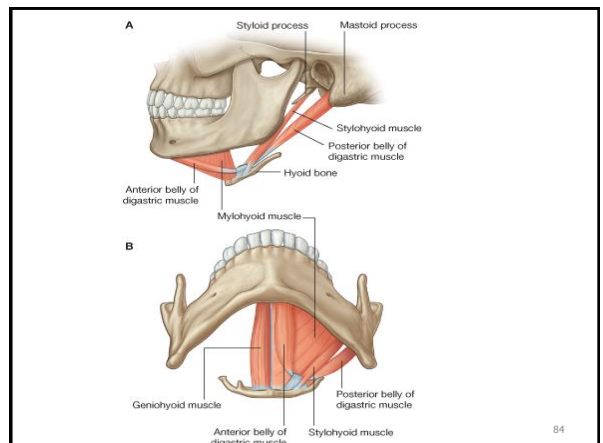
### • Stylohyoid muscle

- Lies above the posterior belly of the digastric muscle
- **Origin** – styloid process
- **Insertion** – body and greater horn of the hyoid
- **Nerve supply** – suprahyoid branch of facial nerve
- **Action** – pull the hyoid bone back & up ward during swallowing

### • Digastric Muscle

- Has 2 bellies connected by tendon
  - **Anterior belly** - digastric fossa of mandible
  - **Posterior belly** – mastoid notch

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## Contents

- **Digastric Muscle...**

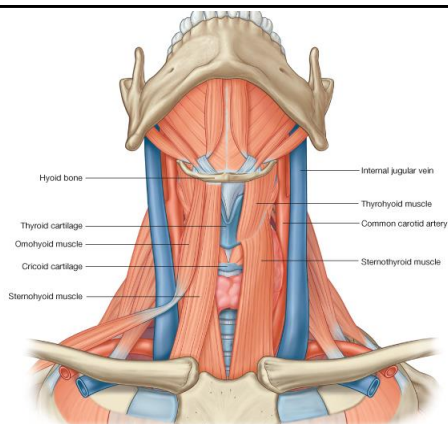
- **Insertion** – intermediate tendon, that passes through fibrous ring, connected to the body and greater horn of the hyoid bone
- **Nerve supply**
  - Anterior belly – **nerve to mylohyoid (branch of mandibular nerve)**
  - Posterior belly – **suprahyoid (branch of facial nerve)**
- **Action** – two bellies elevate hyoid bone; the anterior belly also depress the mandible

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## Infrahyoid Muscles

- Depressors of the larynx and the hyoid bone
- 4 muscles arranged into two layers
- **Superficial layer**
  - Medially - Sternohyoid
  - Laterally- Omohyoid
- **Deep layer**
  - Superiorly - Thyrohyoid
  - Inferiorly- Sternothyroid

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## Infrahyoid Muscles

### Superficial layer

#### 1. Sternohyoid

- **Origin** – posterior surface of the manubrium of sternum and the medial end of clavicle
- **Insertion** – lower border of body of hyoid
- **Innervation** – **ansa cervicalis**
- **Action** – depress hyoid

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## Infrahyoid Muscles

### 2. Omohyoid

- Has two bellies united by tendon connected to clavicle
- **Origin**
  - **Superior**– lower border of body of hyoid bone
  - **Inferior**– suprascapular ligament and notch
- **Insertion**– intermediate tendon which is held in place by loop of deep fascia which connects the tendon to the clavicle and 1st rib
- **Innervation**– **ansa cervicalis**
- **Action**– depress, retract and steadies hyoid

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## Infrahyoid Muscles

### Deep layer

- Superiorly - Thyrohyoid
- Inferiorly- Sternothyroid

### 1. Sternothyroid

- **Origin** – posterior surface of the manubrium of the sternum and the 1st costal cartilage
- **Insertion** – oblique line of lamina of the thyroid cartilage
- **Innervation** – **ansa cervicalis**
- **Action** – depress larynx

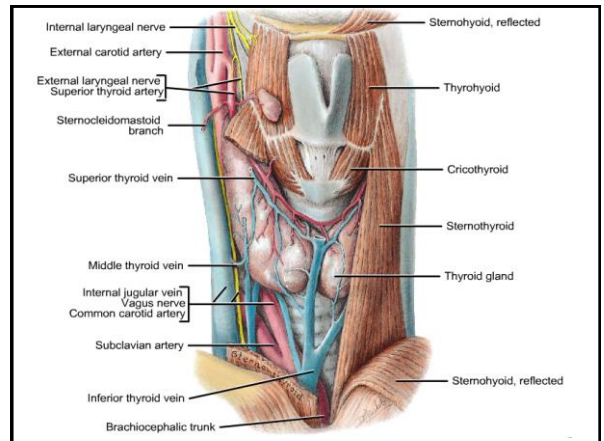
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## Infrahyoid Muscles

### 2. Thyrohyoid

- **Origin** – oblique line of lamina of the thyroid cartilage
- **Insertion** – body and greater horn of the hyoid bone
- **Innervation** – **C1** through hypoglossal nerve
- **Action** – depress hyoid and elevates larynx

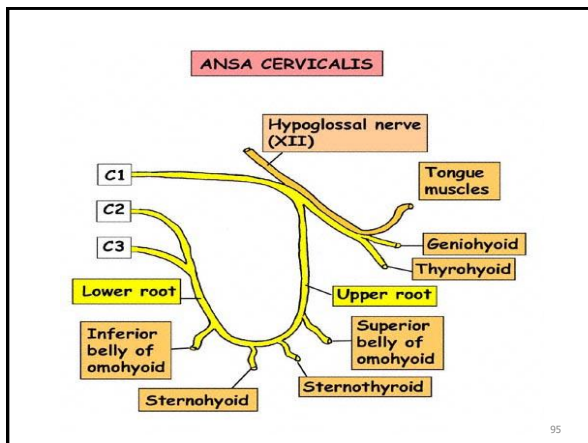
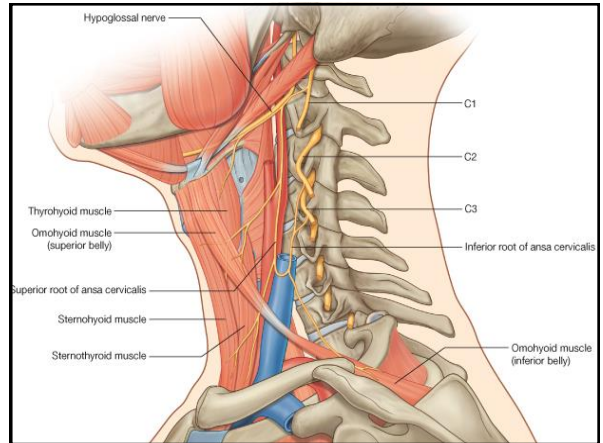
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## The ansa cervicalis

- Nerve loop lies in anterior wall of carotid sheath
- **Formation**
  - Superior root - hypoglossal nerve branch; fibers derived from C1
  - Inferior root – derived from C2 and C3; join superior root in front of CCA
- **Branches to**
  - Superior and inferior belly of omohyoid
  - Sternohyoid
  - Sternothyroid

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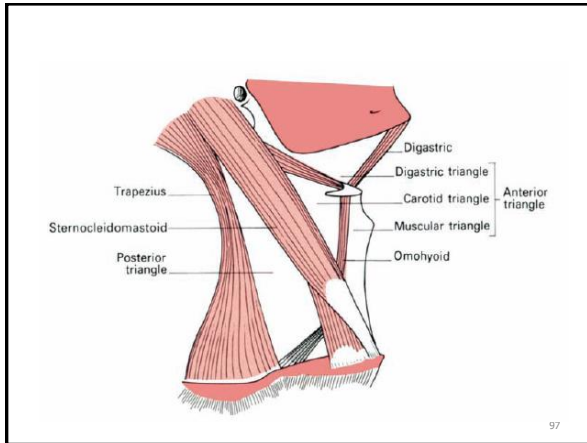


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## Subdivisions of the anterior triangle

- By digastric muscle and superior belly of omohyoid anterior triangle is divided into 4 triangles
  - Submental triangle
  - Submandibular (digastric) triangle
  - Carotid triangle
  - Muscular triangle
- All the triangles except submental triangle are paired

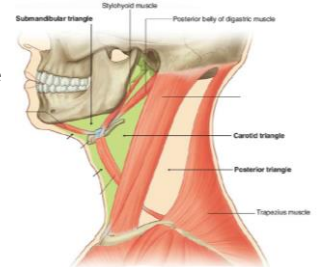
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## Submandibular (digastric) triangle

### ■ Boundaries

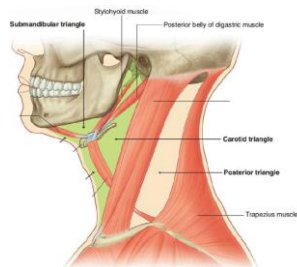
- ✦ **Anteroinferior** – posterior border of anterior bellies of the digastric muscle
- ✦ **Posteroinferior** – anterior border of post. belly of the digastric muscle and Stylohyoid
- ✦ **Superior** - inferior border of the mandible



## Submandibular (digastric) triangle

### ■ Boundaries

- ✦ **Roof** – skin, superficial fascia and deep fascia
- ✦ **Floor** – mylohyoid muscle, Hyoglossus muscle and middle constrictor pharyngeal muscle



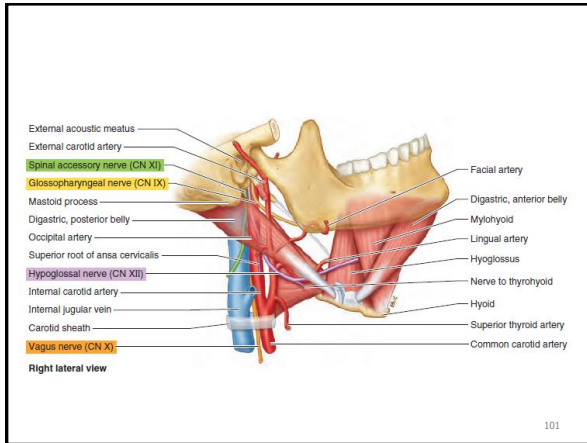
## Contents of digastric triangle

### Anterior part

- ✦ Submandibular salivary gland
- ✦ Submandibular lymph nodes
- ✦ Anterior facial veins
- ✦ Facial artery
- ✦ Hypoglossal nerve
- ✦ Mylohyoid nerve and vessels
- ✦ Submental artery and vein

### Posterior part

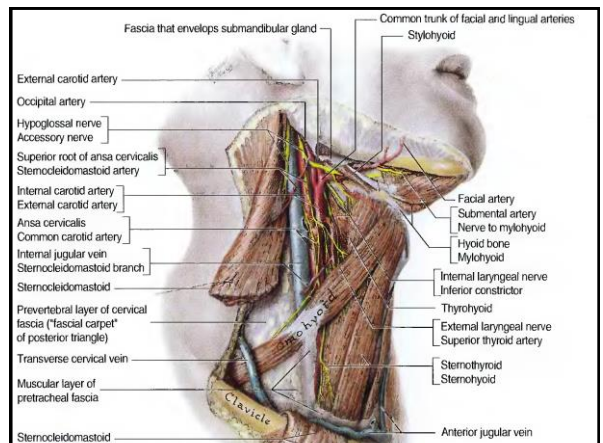
- ✦ Submental artery and vein
- ✦ Carotid arteries
- ✦ Internal jugular vein
- ✦ Glossopharyngeal nerve
- ✦ Vagus nerve
- ✦ Accessory nerves
- ✦ Styloglossus muscle
- ✦ Stylopharyngeus muscle



## Carotid triangle

- **Boundaries**
  - ❖ **Superior** – inferior border of posterior belly of the digastric and Stylohyoid
  - ❖ **Anteroinferior** - upper border of superior belly of the omohyoid
  - ❖ **Posterior** - Anterior border of **SCM**
  - ❖ **Roof** – skin, superficial fascia, deep fascia
  - ❖ **Floor** – thyrohyoid, hyoglossus, middle and inferior constrictors of pharynx

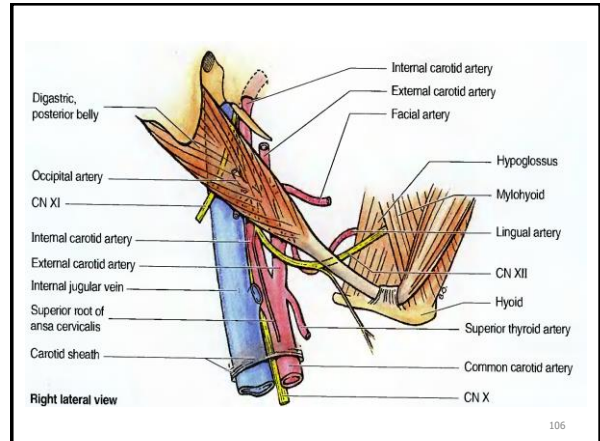
- **Contents**
  - Arteries**
    - Common carotid artery
    - Internal carotid artery
    - External carotid artery
  - Veins** - internal jugular vein and its tributaries
  - Nerves**
    - Vagus & its internal and external laryngeal branches
    - Accessory nerve
    - Hypoglossal nerve
    - Ansa cervicalis
    - Sympathetic trunk
  - **Deep cervical lymph nodes**



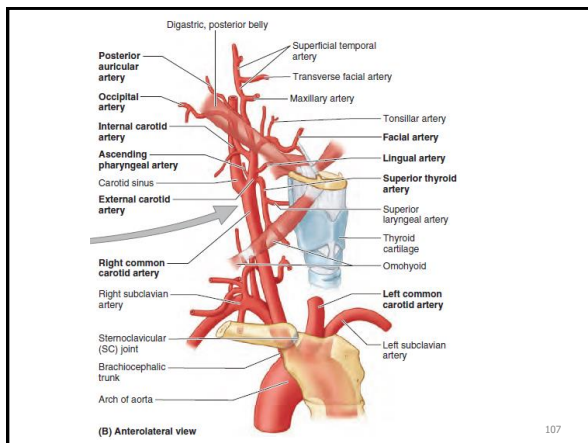
## Carotid sheath

- ✚ Tubular fascial condensation extending from the base of the skull to the root of the neck
- ✚ Contains common carotid artery (medial), internal jugular vein (lateral) and vagus (posterior)
- ✚ at upper part common carotid is replaced by internal carotid
- ✚ Vagus lies between and behind carotid artery and internal jugular vein inside the sheath
- ✚ **Relation**
  - ✓ Ansa cervicalis on anterior wall
  - ✓ Sympathetic trunk is embedded in the posterior wall
  - ✓ deep cervical lymph nodes

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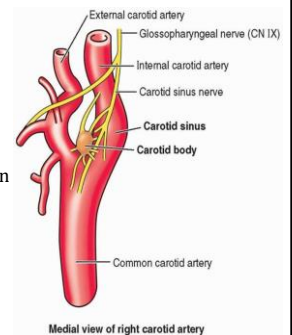
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## Carotid sinus

- ✚ Dilatation of terminal common carotid artery
- ✚ **Proximal part of internal carotid artery**
- ✚ **Innervation** – IX, X and sympathetic nerve
- ✚ **Baroreceptor** – pressure on it may cause fainting

## Carotid body

- ✚ **oval mass** at bifurcation of common carotid artery
- ✚ **Chemoreceptor**  $CO_2$  and  $O_2$
- ✚ **innervation** – same as sinus



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## Muscular triangle

### Boundaries

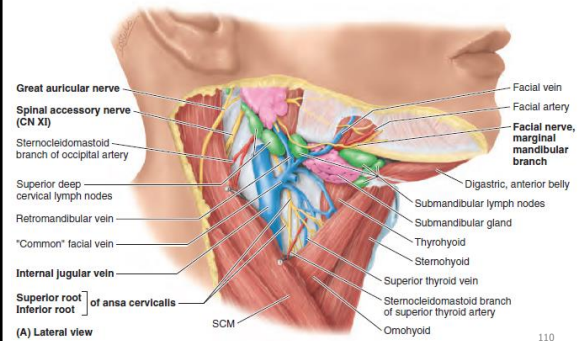
- ✦ **posterosuperior** – superior belly of omohyoid
- ✦ **posteroinferior** - lower anterior margin of the **SCM**
- ✦ **Anterior** - median line of the neck
- ✦ **Floor** – **sternohyoid and sternothyroid muscles**

### Contents

- ✦ **infrahyoid muscle**
- ✦ **thyroid gland, larynx, trachea & esophagus**

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## Muscular triangle



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## Submental triangle (unpaired)

### Boundaries

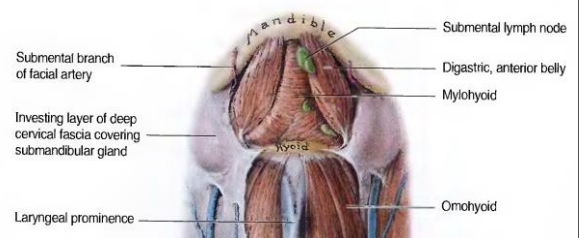
- ✦ **Anterior** - chin
- ✦ **Lateral**- anterior border of right and left digastric muscles
- ✦ **Inferior** – body of hyoid bone
- ✦ **Floor** – mylohyoid muscle

### Contents

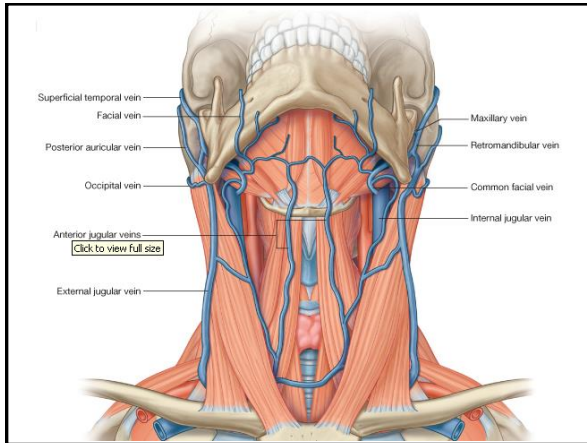
- ✦ **Submental lymph nodes**
- ✦ **Beginning of anterior jugular veins**

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## Submental triangle (unpaired)

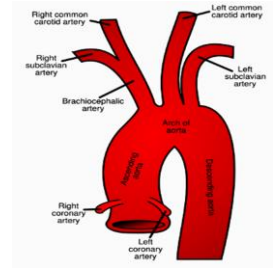


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## Common carotid artery

- ✚ **Right CCA** begins from **brachiocephalic trunk** posterior to sternoclavicular joint
- ✚ **Left CCA** begins from **arch of aorta**
- ✚ Lie within carotid sheath
- ✚ Divides at the level of **superior border of the thyroid cartilage** into internal and external carotid arteries



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## Common carotid artery

### ✚ Relations

- ✓ omohyoid cross it; above it relatively superficial and below it is deeply situated
- ✓ **posteriorly** – transverse process of C5-C7 and muscles attached to them and sympathetic trunk and vessels
- ✓ **medially** – thyroid and neck viscera
- ✓ **laterally** – internal jugular vein

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## Internal carotid

### ▪ Petrous part

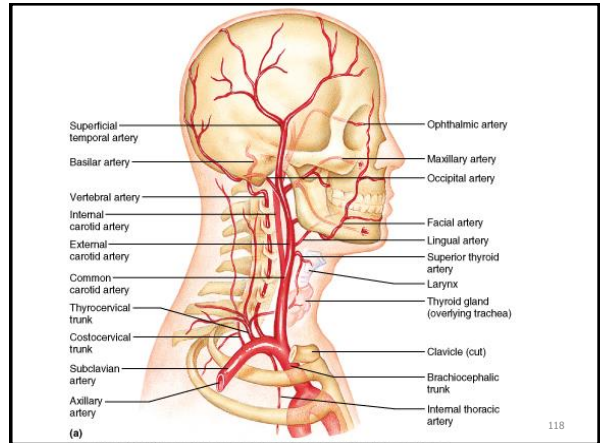
- ✓ in **carotid canal**; course upwards, forwards and medially
- ✓ emerge in posterior wall of **foramen lacerum**
- ✓ surrounded by venous and sympathetic plexuses
- ✓ **Branches** – caroticotympanic and pterygoid

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## External carotid artery

- Begin at superior border of thyroid cartilage
- Lie deep to or superficial to the posterior belly of the digastric
- Anteromedial to internal carotid artery
- **Branches**
  - **Superior thyroid artery** - to the superior pole of the thyroid gland. Its branches are:
    - ✓ Superior laryngeal artery: to the gland
    - ✓ Muscular branch to the SCM muscle & infrahyoid muscle

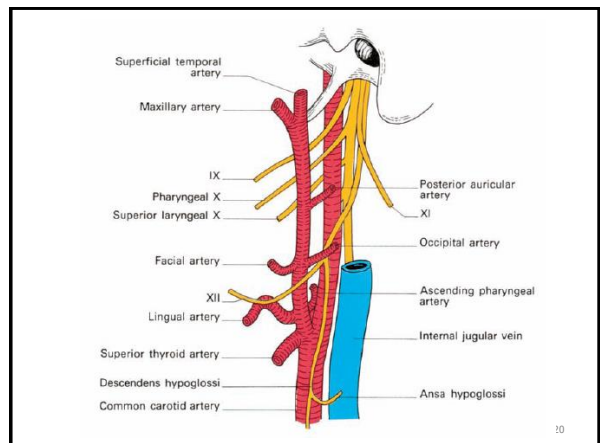
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## External carotid artery

- **Lingual artery** - supplies the tongue
  - ✓ Arise at the middle constrictor, arch superolaterally near greater horn of the hyoid bone
  - ✓ Passes deep to hypoglossal nerve, stylohyoid muscle & posterior belly of digastric muscle
  - ✓ Disappear into the hyoglossus muscle & terminates as deep lingual artery
  - ✓ Gives branch to sublingual gland
- **Ascending pharyngeal artery**
  - ✓ Ascends deep to the internal carotid artery
  - ✓ Branches – pharyngeal, muscular to the prevertebral muscles, middle ear and meninges

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## External carotid artery

### ▪ Facial artery

- ✓ Covered by posterior belly of digastric, stylohyoid muscles & angle of mandible
- ✓ Mark deep groove on submandibular salivary gland before hooking the inferior border of the mandible to join the face (pulse can be felt)
- ✓ **Branches** – tonsillar, palatine and submandibular

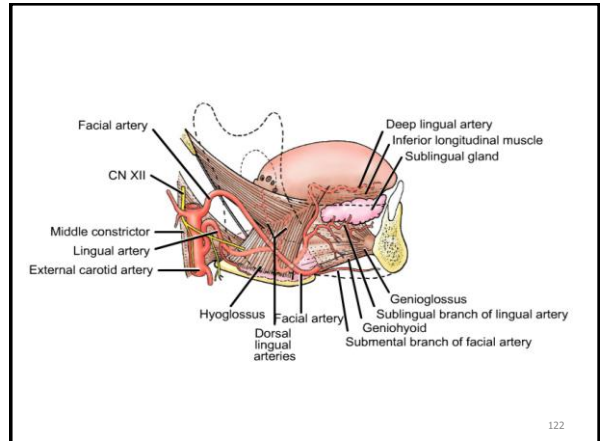
### ▪ Occipital artery

- ✓ Run along the inferior border of the posterior belly of the digastric muscle superficial to the internal carotid, CN IX, X & XI
- ✓ Crosses apex of posterior triangle and ends in the scalp

### ▪ Posterior auricular artery

- ✓ supply back of auricle, mastoid and posterior scalp

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## Internal jugular vein

- Largest vein in the neck
- Drains blood from brain, superficial parts of the face and anterior triangles of the neck
- Course corresponds to a line drawn from a point immediately below the external acoustic meatus to the medial end of clavicle
- Origin – at jugular foramen as continuation of the sigmoid sinus from the cranial cavity

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## Internal jugular vein

### • Course

- inside the carotid sheath lateral to the artery & anterolateral to vagus
- Leave the anterior triangle of the neck through the deep surface of the sternocleidomastoid muscle
- End by forming the brachiocephalic vein with subclavian vein at the sternoclavicular joint

- Right is larger because superior sagittal sinus drains to the right transverse sinus

- Two bulbs are present on the internal jugular vein: superior and inferior

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## Internal jugular vein

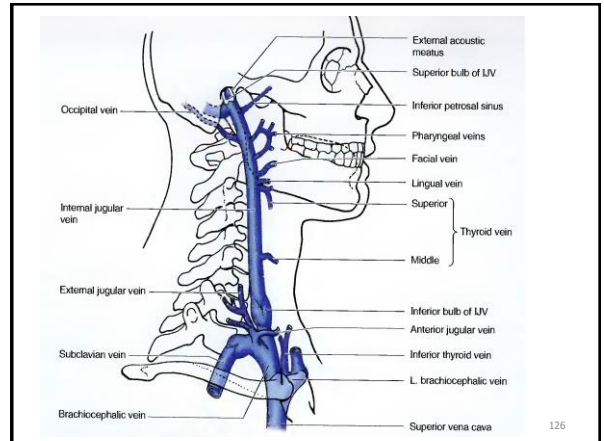
### ▪ Relation

- Superficial – deep cervical lymph nodes
- Posterior – cervical plexus
- Medial – internal and common carotid arteries

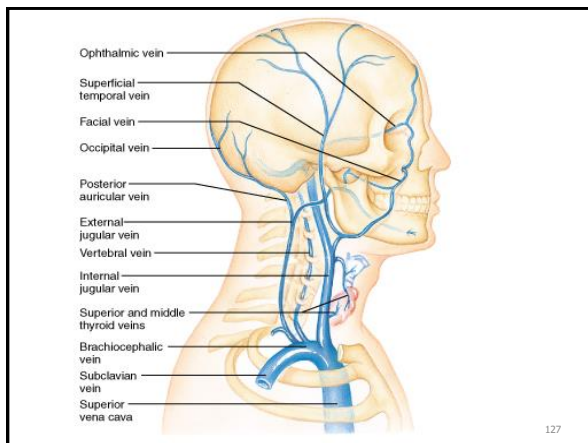
### ▪ Tributaries

- Inferior petrosal sinus, Facial, Lingual, Pharyngeal, Superior & middle thyroid, occipital veins and thoracic duct

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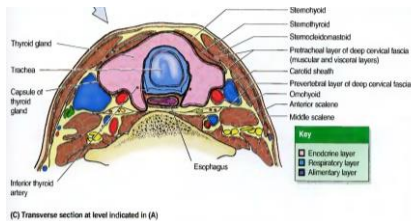
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## Cervical viscera

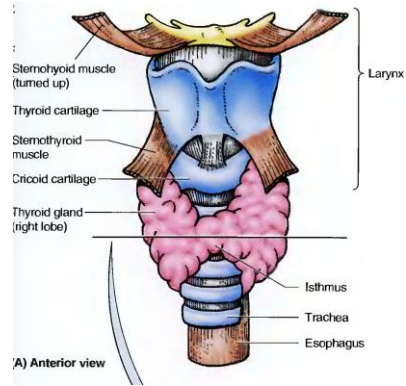
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## Introduction

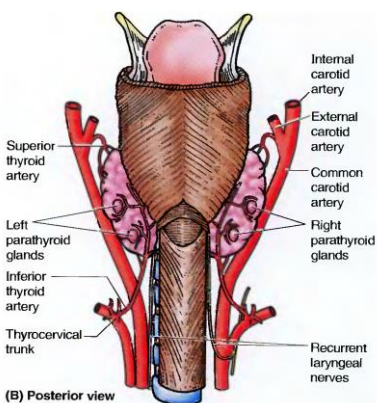
- Disposed into three layers
  - Superficial** – endocrine layer composed of thyroid, parathyroid & thymus
  - Middle** – respiratory layer containing larynx and trachea
  - Deep** – alimentary layer contains esophagus and pharynx



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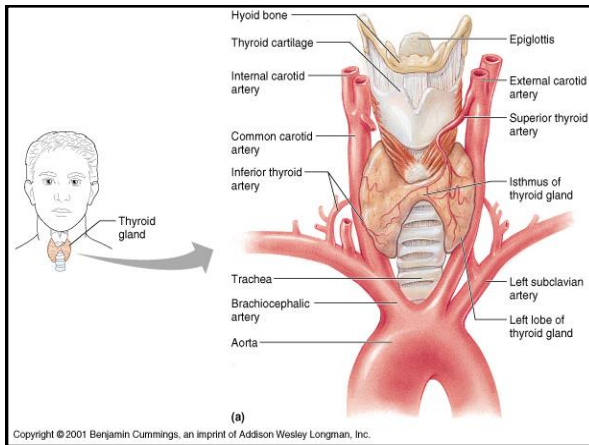


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## Thyroid gland

- 2 lobes lying lateral to the larynx and the trachea; **connected by the isthmus** at the level of tracheal cartilages 2nd and 3rd
- Pyramidal lobe exists in about **half of the people**
  - Small & triangular lobe attached to the upper border of isthmus slightly to the left of the median plane
  - Connected to the hyoid bone** through fibromuscular band called **levator of thyroid gland**

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## Thyroid gland

- **Capsule of the gland – two**
  - **True** – fibrous tissue, continuous with the stroma of the gland
  - **False** – sheath from the pretracheal fascia, attached to larynx, so move during swallowing
  - Between the two capsule – arteries divide and veins form plexus

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## Thyroid gland

### Relations

- **Isthmus**
  - **Posteriorly** – 2nd & 3rd tracheal rings
  - **Anteriorly** – skin + fascia, sternothyroid & sternohyoid muscles, anterior jugular vein
  - **Inferiorly** – inferior thyroid vein
- **Lobes**
  - Conical in shape, lies opposite to C5-7 & T1
  - Each lobe has three surfaces, 2 borders, apex and base

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## Thyroid gland

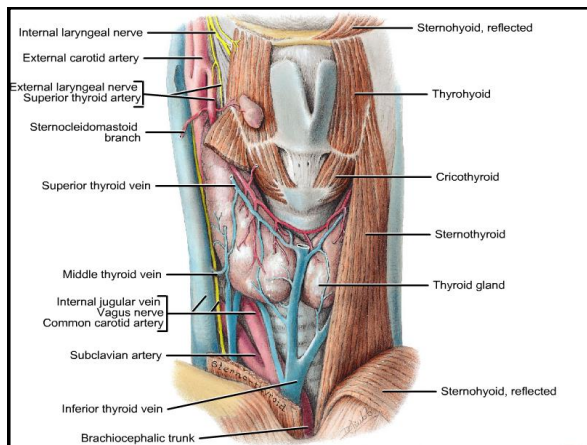
- **Apex**
  - reach up to the oblique line of the lamina of the thyroid cartilage
- **Base**
  - up to 4th and 5th rings of cartilage
- **Lateral (superficial) surface**
  - related to sternothyroid, sternohyoid, superior belly of omohyoid & anterior border of the sternocleidomastoid
- **Medial surface**
  - related to trachea, esophagus, inferior constrictor muscle, cricothyroid muscle, external and recurrent laryngeal nerves
- **Posterolateral surface**
  - carotid sheath

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## Thyroid gland

- **Anterior border**
  - anterior branch of superior thyroid artery
- **Posterior border**
  - inferior thyroid artery, anastomoses between inferior and superior thyroid arteries, parathyroid gland and thoracic duct

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## Thyroid gland

### Arterial supply

- **Superior thyroid artery**
  - A branch of external carotid artery
  - Runs downwards and forwards accompanied by external laryngeal nerve
  - At upper pole divide into anterior and posterior branches
  - Anterior branch descends on anterior border and along upper border of isthmus to anastomose with the other
  - Posterior branch descends on posterior border and anastomose with ascending branch of inferior thyroid artery, supply parathyroid gland

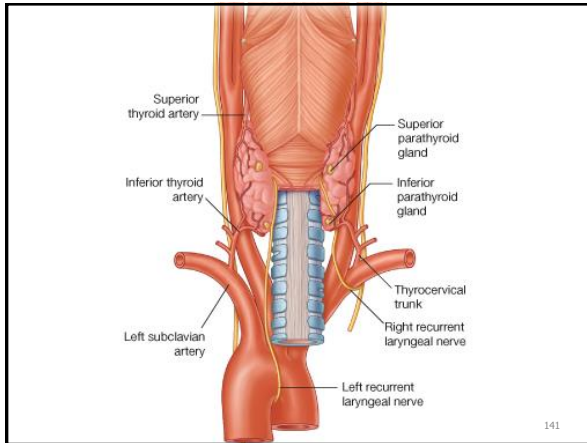
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## Thyroid gland

### ▪ Inferior thyroid artery

- A branch of thyrocervical trunk
- Runs upwards and medially and downwards to lower pole
- Divided into 4 or 5 glandular branches
- Terminal part related to recurrent laryngeal nerve
- Ascending branch anastomoses with posterior branch of superior thyroid
- ✓ Superior thyroid artery supplies upper 1/3 of lobes and upper half of isthmus and inferior thyroid supplies the rest; but they anastomoses freely

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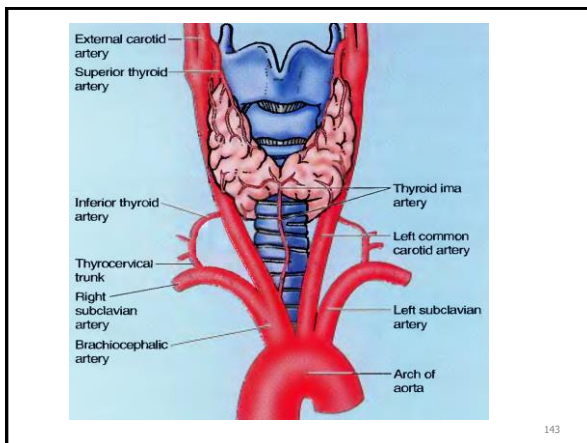
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## Thyroid gland

### ■ Thyroid ima artery (lowest thyroid artery)

- Variable, in approximately 10% of people, a small, unpaired
- Usually arises from brachiocephalic trunk
- May arise from the arch of aorta or from the right common carotid, subclavian, or internal thoracic arteries
- Ascends on the anterior surface of the trachea, which it supplies, and continues to the isthmus of the thyroid gland, where it divides into branches that supply it
- The possible presence of a thyroid ima artery must be considered when performing procedures in the midline of the neck inferior to the isthmus (e.g., for a tracheostomy)
  - As it runs anterior to the trachea, it is a potential source of bleeding

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## Thyroid gland

### ■ Venous drainage

- **Superior thyroid vein** – emerge at upper pole and ends in internal jugular vein or common facial vein
- **Middle thyroid veins** – emerge at middle of the lobe and ends in internal jugular vein
- **Inferior thyroid vein** – unpaired; emerge at lower border of isthmus, form plexus in front of trachea and ends in internal jugular vein or left brachiocephalic vein

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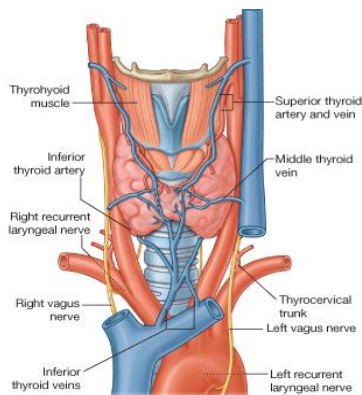
## Thyroid gland

### ▪ Lymphatic drainage

- **Upper part** - drains to upper deep cervical lymph nodes
- **Lower part** - drains to lower deep cervical lymph nodes

- **Nerve supply** - from the 3 cervical ganglia, reach through cardiac and laryngeal branch of vagus

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## Applied anatomy

### ▪ Goiter

- Enlargement of the gland
- Exerts pressure on trachea and recurrent laryngeal nerve

### ▪ Thyroidectomy

- Removal of the gland
- Posterior part is preserved to avoid removal of parathyroids
- Superior and inferior thyroid arteries are ligated to avoid damage to external and recurrent laryngeal nerves
- Inferior thyroid veins and thyroidea ima artery are sources of bleeding

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## Parathyroid glands

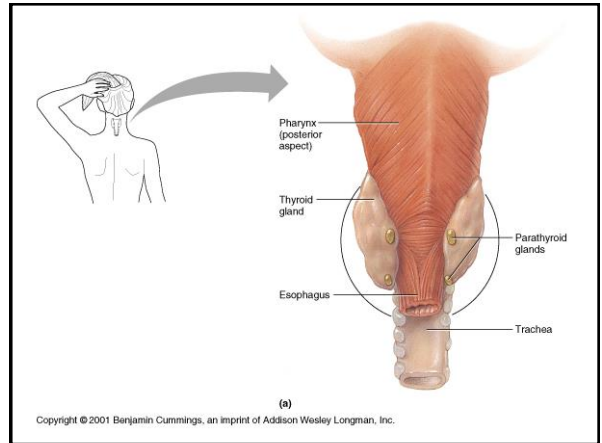
- Small & pea shaped
- Two in each lobe; superior & inferior
- Embedded in the posterior wall of the fibrous capsule of thyroid gland
- Superior one is more constant, present near middle of posterior surface
- Inferior is more variable, near or below the inferior pole

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## Parathyroid glands

- **Arterial supply** – branches from anastomoses between superior & inferior thyroid arteries
- **Venous drainage** – plexus of veins on the anterior surface of thyroid gland and trachea
- **Lymphatic drainage** – with thyroid and thymus into deep cervical lymph nodes and paratracheal groups
- **Nerve supply** – from middle and superior cervical ganglia

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## Trachea (cervical portion)

- Non collapsible fibrocartilagenous tube, extends from larynx to lung roots
- Kept patent by C- shaped cartilaginous rings; the posterior deficient part is covered by trachealis muscle and fibrous tissue – allows for expansion of esophagus
- Whole length = 12 cm; lower level of cricoid cartilage (C6) to level of T4/T5
- Descends in the midline of the neck as a continuation of the larynx

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## Trachea (cervical portion)

### Relations

- **Anteriorly**
  - Isthmus of thyroid gland in front of 2nd & 3rd rings
  - Inferior thyroid vein
  - Thyroid ima (if present)
  - Pretracheal fascia
  - Sternohyoid
  - Sternothyroid
- **At the root of the neck**
  - Left innominate vein
  - Jugular venous arch

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## Trachea (cervical portion)

### Relations

- **Laterally**
  - Lobes of thyroid gland
  - Carotid sheath and its content
- **Posteriorly**
  - Esophagus
  - Recurrent laryngeal nerve

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## Trachea (cervical portion)

- **Arterial supply** - branch from inferior thyroid artery
- **Veins** – inferior thyroid vein
- **Lymph vessels** – pretracheal and paratracheal
- **Nerve supply**
  - Parasympathetic - recurrent laryngeal
  - Sympathetic – cervical ganglion

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## Applied anatomy

- **Tracheotomy**
  - surgical incision through anterior wall of trachea in case of laryngeal obstruction
  - done in 2nd and 3rd rings (covered by isthmus) because supra isthmus is liable to stricture and infra isthmus is dangerous due to thyroid vessels

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## Esophagus (cervical portion)

- Distensible muscular tube from pharynx to stomach
- Begins as a continuation of pharynx at the level of **cricoid cartilage**
- Ends in the abdomen opposite to T11 by joining the cardiac orifice of the stomach
- Whole length = 25cm with 3 parts; cervical, thoracic & abdominal
- Course – begin in the median plane then turns slightly to the left as it descends in the neck

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## Esophagus (cervical portion)

### ▪ Relations

- **Anteriorly** – trachea and recurrent laryngeal nerve
- **Posteriorly** – longus coli and vertebral column
- **Laterally** – lobe of thyroid gland, carotid sheath & its content
- **At the root of the neck**
  - To the right side – cervical pleura
  - To the left side – subclavian artery and thoracic duct

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## Esophagus (cervical portion)

- **Arterial supply** - branches from inferior thyroid artery
- **Veins** – inferior thyroid vein
- **Lymph vessels** – deep cervical lymph nodes
- **Nerve supply** – recurrent laryngeal & sympathetic trunk

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## Pharynx

- Funnel shaped fibromuscular tube found posterior nasal, oral and laryngeal cavities
- Extends from the **base of skull to inferior border of cricoid cartilage (C6)** = 12 cm long
- Lies between the bodies of the vertebrae and the larynx between C4-6.
- Becomes continuous with the esophagus at C6
- Conducts food and air
- The cavity above the inlet of larynx is wide and always open
- The cavity below the inlet of the larynx is narrow, the anterior & posterior parts are in contact except during food passes

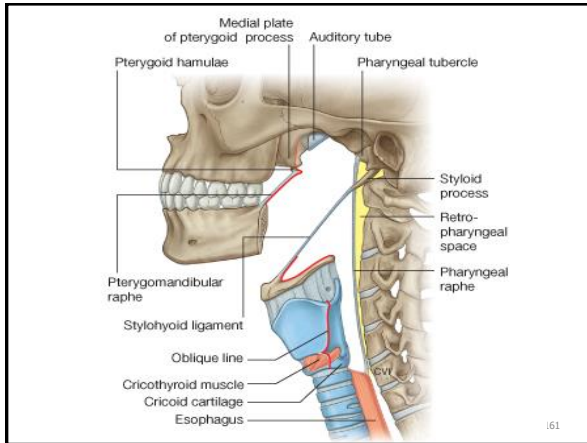
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## Pharynx

### ▪ Boundaries and relation

- **Superior** – body of sphenoid and basilar part of occipital
- **Inferior** – continuous with esophagus
- **Posterior** – prevertebral fascia
- **Anterior** – communicate with nasal cavity, oral cavity and larynx
- **Lateral** – attached to medial pterygoid plate, pterygomandibular raphe, mandible, tongue, hyoid, thyroid and cricoid cartilage

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## Pharynx

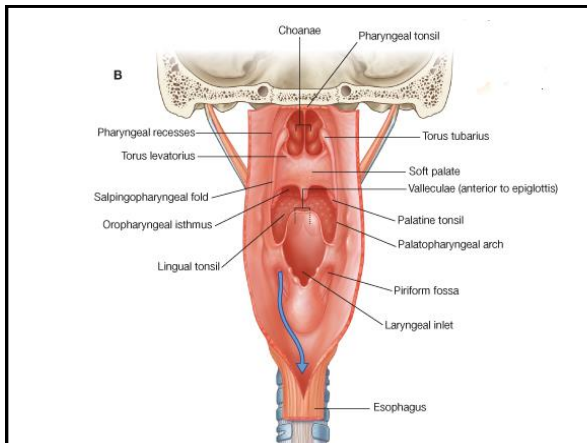
### ▪ Openings in the pharynx

#### – Anterior

- Two posterior nasal openings
- Oropharyngeal isthmus
- Inlet of larynx

#### – Lateral – openings of the auditory tube

#### – Inferiorly – into esophagus



## Pharynx

### ▪ Pharyngobasilar fascia

- Covers the internal surface of the constrictor muscles
- Connects pharynx with the base of the skull and fill the gap between the superior constrictor and skull
- Together with the buccopharyngeal fascia fill the gaps in the muscular wall

### ▪ Buccopharyngeal fascia

- covers the external surface of the buccinator muscle and the superior constrictor muscle
- contains: pharyngeal venous plexus and pharyngeal plexus of nerves

## Pharynx

### ▪ The pharyngeal plexus of veins

- lies deep to this fascia, draining the pharynx, soft palate and the pharyngeal tonsil
- communicates with the pterygoid plexus of veins and drains into the internal jugular vein
- These veins have no valves and infection from the pharynx, palate and tonsils may spread into the meningeal venous plexus

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## Pharynx

### ▪ The pharyngeal nervous plexus

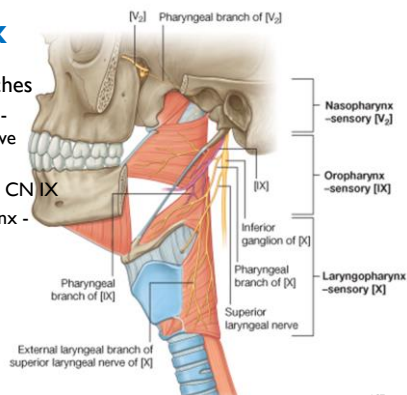
- formed by branches from CN IX, X and superior cervical ganglion
- Motor branches – CN X
  - supply all muscles of pharynx and palate except **stylopharyngeus (CN IX)** and **tensor veli palati (CN V3)**
  - inferior constrictor receive additional fibers from **external and recurrent laryngeal nerves**

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## Pharynx

### ▪ Sensory branches

- nasopharynx - maxillary nerve (CN V2)
- oropharynx - CN IX
- laryngopharynx - vagus



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## Pharynx

### ▪ Muscular layer

- Inner longitudinal and outer circular skeletal muscles
- 6 paired skeletal muscles
  - **Three pairs of circular muscles** (external muscles): superior, middle and inferior constrictor muscles
  - **Three pairs of longitudinal** (internal muscles): palatopharyngeus, stylopharyngeus & salpingopharyngeus

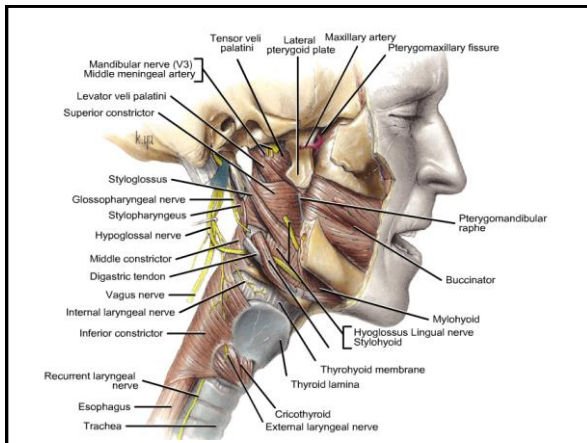
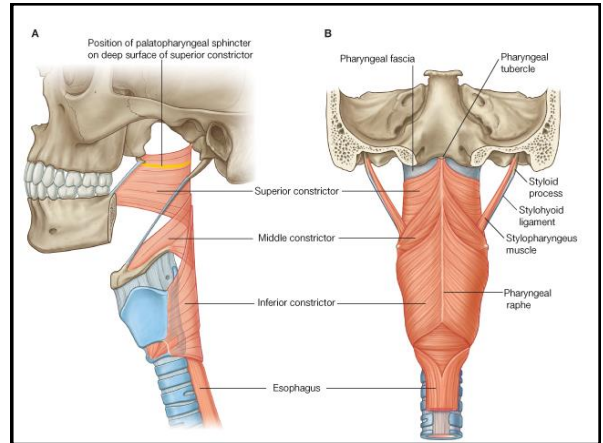
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## Pharynx

### ▪ External muscles

- Originate anteriorly and pass posteriorly and meet in midline fibrous raphe
- Inferior overlap middle and middle overlap superior
- Constrict pharynx during swallowing

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## Pharynx

### ▪ Superior constrictor muscle

- lies behind the nasal and oral cavities and continuous with the plane of the buccinator
- **Origin** – pterygoid hamulus, pterygomandibular raphe, posterior end of mylohyoid line of mandible & the side of the tongue
- **Insertion** – median raphe of pharynx and pharyngeal tubercle

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## Pharynx

### ▪ Middle constrictor muscle

- lies behind the tongue and larynx
- It is truly a constrictor and plays a major role in swallowing
- **origin** - stylohyoid ligament, greater and lesser horns of hyoid bone
- **Insertion** - median raphe of pharynx

### ▪ Inferior constrictor muscle

- **Origin** – oblique line of thyroid cartilage and the sides of cricoid cartilage
- **Insertion** – median raphe of pharynx
- The most inferior aspect is the **cricopharyngeus** which is the **sphincter b/n the pharynx and the esophagus**

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## Pharynx

### ▪ Innervation of constrictor muscles

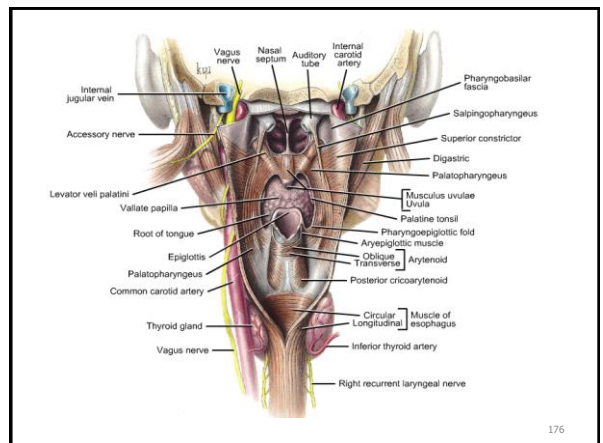
- **Superior and middle constrictor** muscles are innervated by
  - ✓ the pharyngeal branches of the vagus nerve
- **Inferior constrictor** muscle and the **cricopharyngeus** are innervated by
  - ✓ recurrent branches of the vagus nerve

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## Pharynx

- The three muscles overlap each other and form four gaps for structures to enter or leave the pharynx
  - The interval between the superior constrictor and base of the skull: allows for
    - ✓ Levator veli palatini
    - ✓ auditory tube
    - ✓ ascending palatine artery

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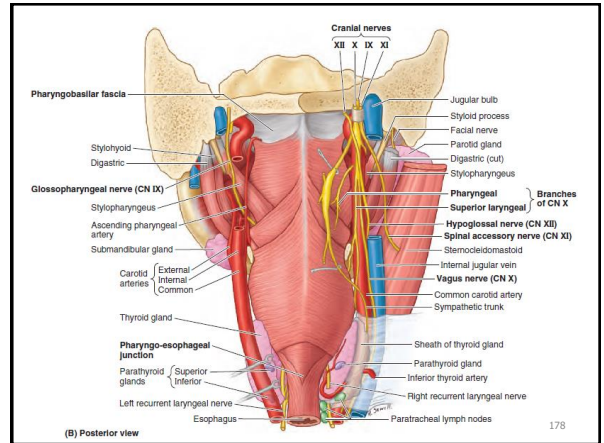


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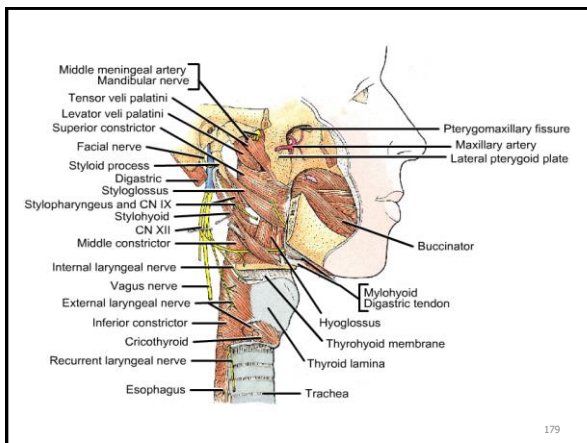
## Pharynx

- The interval between superior and middle constrictor
  - ✓ stylopharyngeus
  - ✓ cranial nerve IX
  - ✓ Stylohyoid ligament
- The interval between middle & inferior constrictors covered by **the thyrohyoid membrane** transmits
  - ✓ Internal laryngeal nerve
  - ✓ Superior laryngeal artery into the larynx
- Between the inferior constrictor and esophagus
  - ✓ Inferior laryngeal vessels
  - ✓ Recurrent laryngeal nerve

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## Pharynx

- **Internal muscles of the pharynx**
  - Elevates larynx and pharynx during swallowing
- **Stylopharyngeus**
  - **Origin** - styloid process of temporal bone
  - **Insertion** - posterior & superior border of the thyroid cartilage
  - **Course** - descends between external & internal carotid arteries and enters the pharyngeal cavity with IX between the superior and middle constrictor muscles
  - **Nerve supply** - CN IX

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## Pharynx

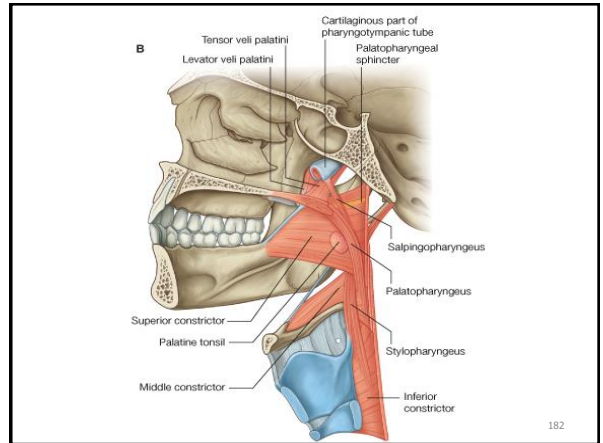
### Palatopharyngeus

- **Origin** - hard palate and palatine aponeurosis
- **Insertion** – posterior border of lamina of thyroid cartilage and sides of pharynx
- **Nerve supply** – pharyngeal plexus
- The overlying mucosa forms the palatopharyngeal arch, posterior to the palatine tonsil

### Salpingopharyngeus

- **Origin** – cartilaginous part of auditory tube
- **Insertion** – blend with palatopharyngeus
- **Nerve supply** - pharyngeal plexus

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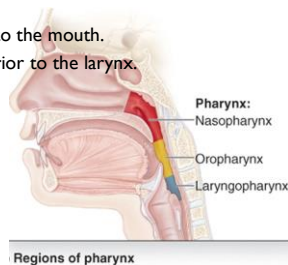


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## Pharynx

- **Interior of Pharynx.** The pharynx is divided into three parts:

- ✓ **Nasopharynx:** posterior to the nose and superior to the soft palate.
- ✓ **Oropharynx:** posterior to the mouth.
- ✓ **Laryngopharynx:** posterior to the larynx.



Regions of pharynx

## Pharynx

### Nasopharynx

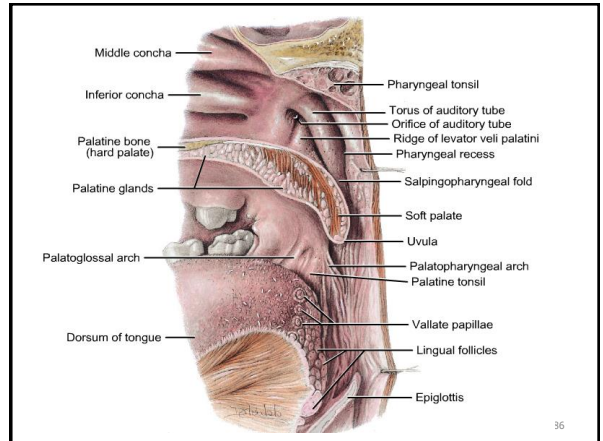
- it is the posterior extension of the nasal cavities above the soft palate
- Respiratory function
- **Walls**
  - **Roof** – sphenoid & occipital bone, mucus membrane, submucosa (contain pharyngeal tonsil)
  - **Floor** – soft palate
  - **Anterior wall** – two posterior openings of the nasal cavity
  - **Posterior wall** - anterior part of atlas lined with mucus membrane

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## Pharynx

- **Lateral wall** – presents
  - Pharyngeal openings of **auditory tube** – behind inferior nasal concha
  - Tubal elevation- posterior to the auditory openings
  - **Salpingopharyngeus fold** – inferior to tubal elevation due to the muscle
  - **Pharyngeal recess** – depression behind tubal elevation
  - **Tubal tonsil** – lymphoid tissue in the submucosa behind the openings of the auditory tube
- **Inferiorly** – Nasopharynx communicate with oropharynx at nasopharyngeal isthmus

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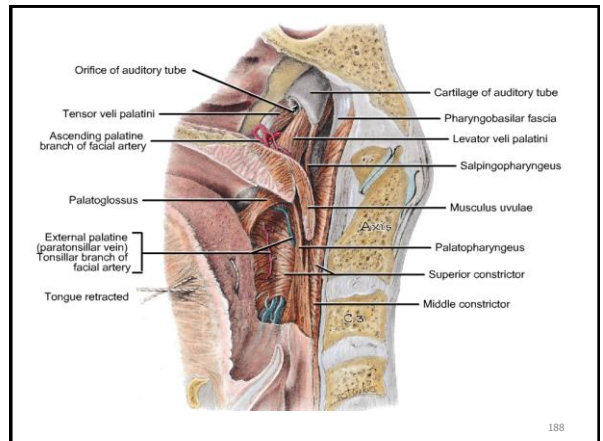
## Oropharynx

- ✓ Behind oral cavity, extends from soft palate to the upper border of epiglottis

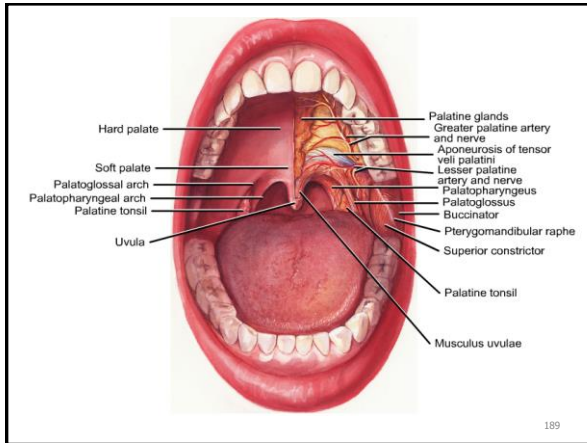
### Walls

- **Anterior**
  - Communicate with oral cavity through oropharyngeal isthmus (isthmus of fauces )
    - ✓ Laterally bounded by two arches – palatoglossal & palatopharyngeal; between them triangular fossa for **palatine tonsil**
    - ✓ Inferiorly pharyngeal part of tongue – contain **lingual tonsil**.
- **Posterior wall** – body of axis and upper part of C3 lined with mucus membrane

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- Tonsillar ring (Waldeyer's ring) of lymphoid tissue
  - In relation to oropharyngeal isthmus, form circular band of lymphoid tissue
  - Formed by palatine, lingual, pharyngeal and tubal tonsils
  - Forms defense system to entry of lower respiratory organs

## ▪ Laryngopharynx

- Found posterior to the larynx
- Extend from superior border of epiglottis to inferior border of cricoid.

### Walls

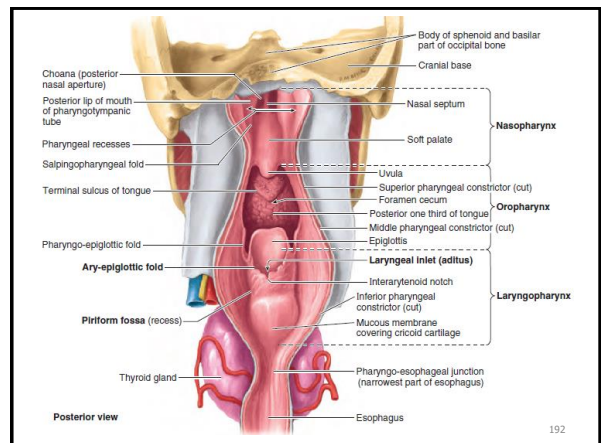
#### – Anterior

- Communicate with larynx through inlet of larynx; bounded by epiglottis, aryepiglottic fold & inter arytenoid folds

- Mucous membrane covering posterior surface of larynx

#### – Posterior – related to C4 - C6 vertebrae

- Its posterior and lateral walls are formed by the middle and inferior pharyngeal constrictor muscles



## ▪ Laryngopharynx

- Piriform recess (fossa):
  - is a small depression of the **laryngopharyngeal cavity** on either side of the laryngeal inlet.
- **Bounded by**
  - ✓ aryepiglottic fold medially &
  - ✓ thyroid cartilage and thyrohyoid membrane laterally
- site for arrest of foreign body and produce cough
- the mucosa is supplied by **internal laryngeal nerve**
- Branches of internal and recurrent laryngeal nerves lie in the recess

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## Larynx

### ▪ Functions

- Voice production
- Air passage to trachea
- Acts as a valve for preventing swallowed food from entering the lower respiratory tract

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## Larynx

### ▪ Position

- ✓ At midline of neck from root of tongue to trachea
- ✓ In front of laryngopharynx opposite C3 to C6
- ✓ Superiorly continuous with laryngopharynx through the inlet of larynx
- ✓ Inferiorly continuous with trachea
- ✓ 5 cm in adult man but slightly shorter in female & children

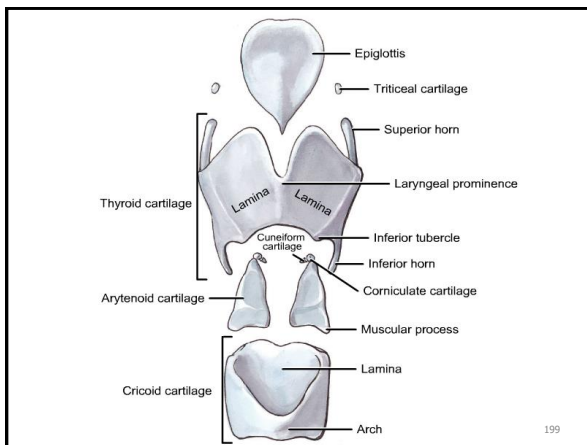
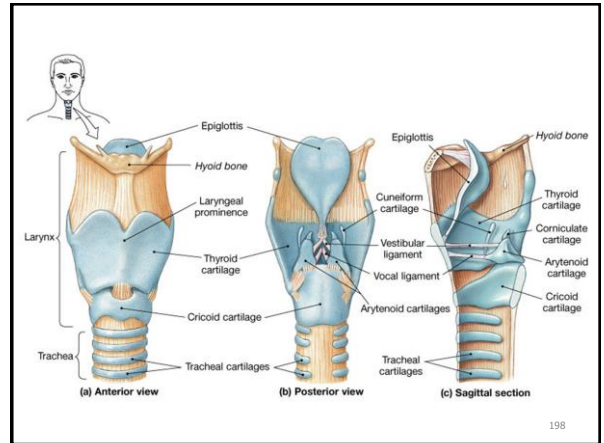
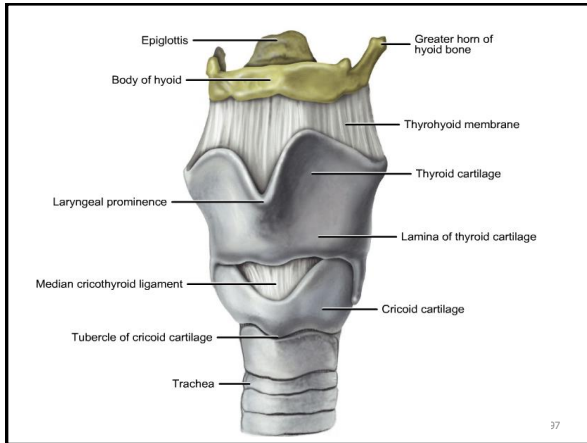
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## Larynx

### ▪ Structures

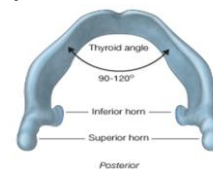
- 9 cartilages joined by ligaments and membranes
  - Three unpaired – thyroid, cricoid & epiglottis
  - Three paired – arytenoid, corniculate & cuneiforms
- Muscles

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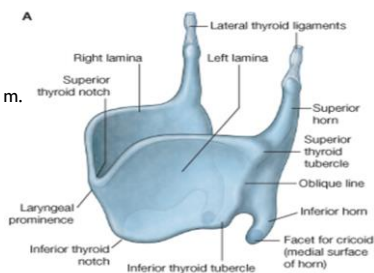
### ■ Thyroid cartilage

- ✓ Made up of two quadrilateral plates called laminae
- ✓ Placed obliquely; where the inferior 2/3 is fused anteriorly in the midline
- ✓ The angle of fusion is **more acute** in the male and forms the **laryngeal prominence (Adam's apple)** after puberty
- ✓ Superior to the laryngeal prominence there is a V shaped notch called **thyroid notch**



## ▪ Thyroid cartilage

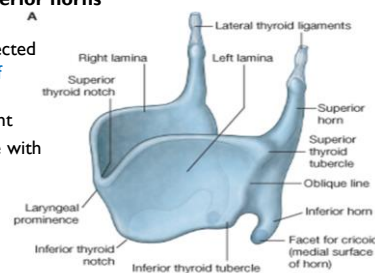
- ✓ The upper border is attached to the **hyoid bone** by the **thyrohyoid membrane**
- ✓ Lateral surface is marked by an **oblique line** for the attachment of
  - sternothyroid,
  - thyrohyoid and
  - Inf. constrictor m.



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## ▪ Thyroid cartilage

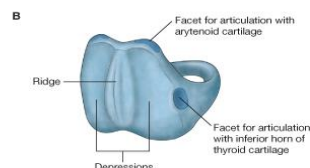
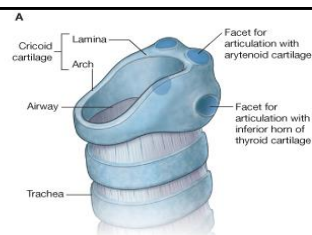
- ✓ Each lamina has projection from posterior border called **superior and inferior horns**
- Sup. horn is connected to **greater horn of hyoid** by lateral thyrohyoid ligament
- Inf. horn articulate with **cricoid cartilage**



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## ▪ The cricoid cartilage

- ✓ Ring shaped, having quadrilateral lamina posteriorly and narrow arch anteriorly
- ✓ its anterior arch is attached superiorly to the thyroid cartilage by the **cricothyroid membrane**
- ✓ The thickened midline portion of this membrane is the median **cricothyroid ligament**
- ✓ Lamina projects upwards and articulates with **arytenoid**

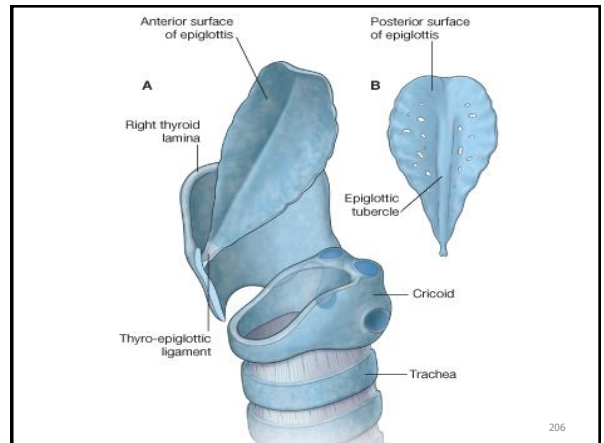


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### ■ The epiglottis

- ✓ leaf-shaped, broad flat plate of elastic cartilage at anterior wall of upper part of larynx
- ✓ **upper end** – broad, free, projects behind the tongue and hyoid bone, above and in front of the inlet of larynx
- ✓ **lower end** – pointed, attached above the angle of fusion of the thyroid cartilage by the **thyroepiglottic ligament**
- ✓ anterior surface is connected to tongue by **glossoepiglottic fold** and hyoid by **hyoepiglottic ligament**

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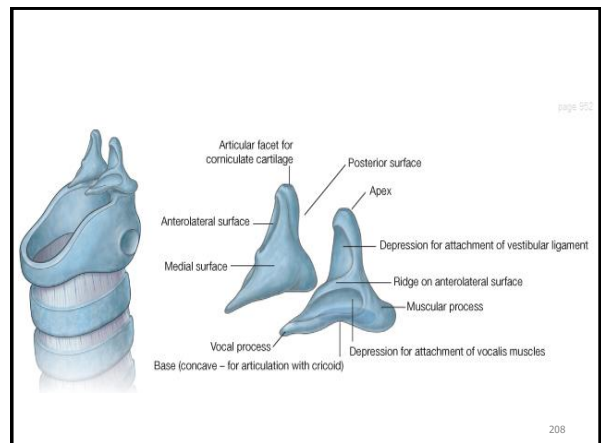


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### ■ The arytenoid cartilages

- ✓ Three sided pyramid with a base and apex
- ✓ On upper border of lamina of cricoid
- ✓ **Apex** – curved posteromedially and articulate with **corniculate cartilage**
- ✓ **Base** - articulate with cricoid lamina through synovial joint called **crico-arytenoid**
- ✓ **Has two processes**
  - vocal process - anterior & attach to the vocal ligament
  - muscular process- lateral & serves as attachment points for muscles of the larynx
- ✓ Control the opening between the **vocal cords**, the **rima glottidis**

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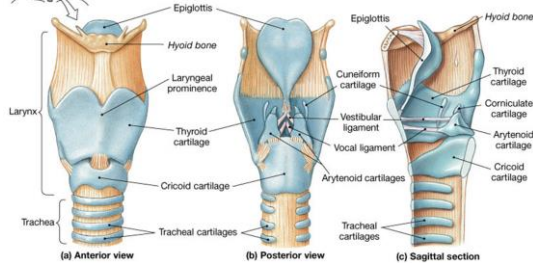
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### ▪ Corniculate

- ✓ Small conical nodules of fibrocartilage articulating with
  - the apex of arytenoid cartilage in the posterior part of aryepiglottic fold

### ▪ Cuneiform; Rod shaped, found in the aryepiglottic fold, ant to the arytenoid cartilage



## Joints of larynx

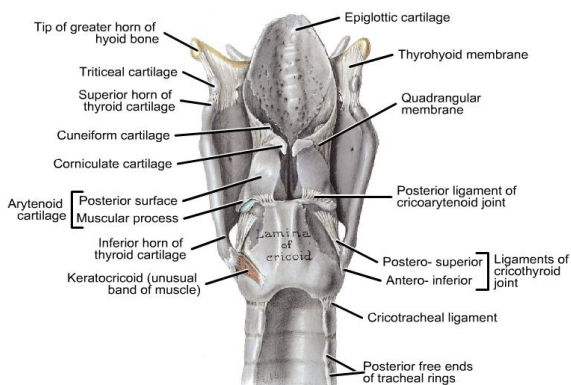
### ▪ Crico-arytenoid

- ✓ Synovial, between base of arytenoid and upper border of lamina of cricoid cartilage
- ✓ Permits **sliding, tilting and rotary movements** of arytenoids which are important in approximating, tensing and relaxing vocal folds

### ▪ Crico-thyroid

- ✓ Synovial, between **inferior horn of thyroid cartilage** and the sides of **cricoid cartilage** at the junction of its lamina and arch
- ✓ Permits rotary and gliding movements which permit changes in length of vocal folds

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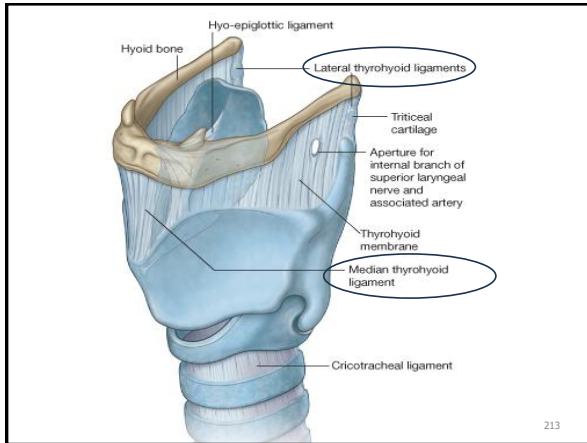


## Ligaments and membranes

### ▪ Thyrohyoid membrane & ligament

- ✓ Connect upper border of thyroid cartilage to body, greater & lesser horn of hyoid
- ✓ **Midline thickening – median thyrohyoid ligament**
- ✓ **Posterior border thickening – lateral thyrohyoid ligament**
- ✓ Outer surface covered with thyrohyoid muscle
- ✓ **Pierced** by superior laryngeal vessels & internal laryngeal nerve

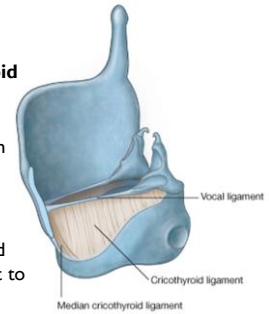
212



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#### ■ Cricothyroid ligament

- ✓ Triangular in shape
- ✓ Connects upper border of **arch of cricoid cartilage** with deep surface of **thyroid cartilage** near the lower border
- ✓ Outer surface covered with **cricothyroid muscle**
- ✓ Inner surface covered with **mucous membrane**
- ✓ Soft spot inferior to thyroid cartilage - air way is closest to skin and most accessible



214

#### ■ Vocal ligament, vocal fold & conus elasticus

##### ✓ Conus elasticus

- triangular ligament connecting cricoid, thyroid & arytenoid cartilage

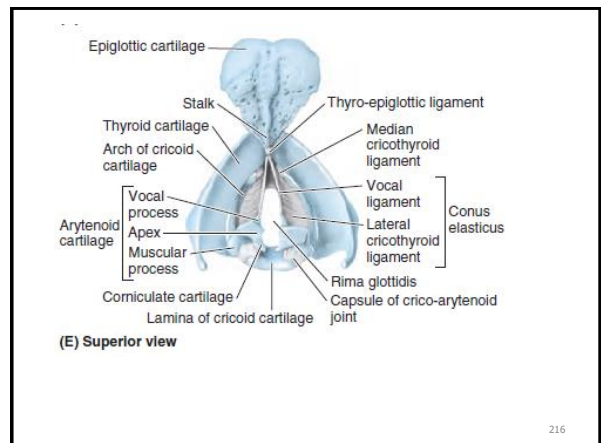
##### ✓ Vocal ligaments

- attach between the vocal processes of the arytenoid cartilages and the thyroid cartilage, behind its fusion angle
- i.e. thickened upper border of conus elasticus
- **forms skeleton of vocal folds**

##### ✓ Vocal fold

- the upper border of conus elasticus covered with the mucous membrane

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216

## ▪ Quadrate membrane & vestibular ligaments

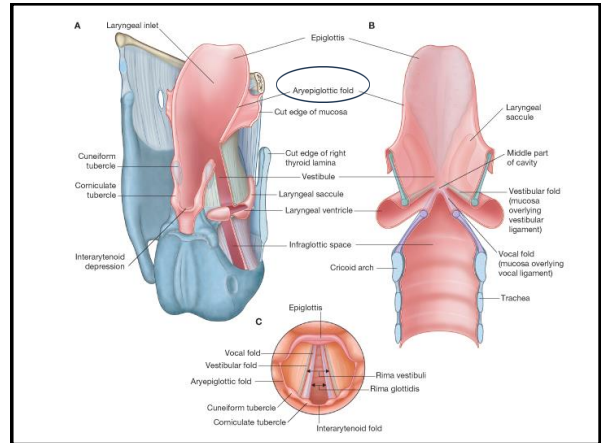
### ✓ Quadrate membrane

- Thin submucosal sheath extending from arytenoid cartilage to the epiglottis
- Superior border **forms aryepiglottic fold**
- Inferior border forms vestibular fold

### ✓ Vestibular ligaments

- The free inferior margin the quadrate membrane
- Covered with mucous membrane called vestibular fold (superior to the vocal fold)
- Extends from the **arytenoid to the thyroid cartilage**

217



## ▪ Hyoepiglottic ligament

- ✓ Connects the anterior surface of the epiglottis to the back of hyoid

## ▪ Thyro-epiglottic ligament

- ✓ Lower end of the epiglottis to the back of thyroid

## ▪ Ary-epiglottic folds of mucus membrane

- ✓ Connects the sides of the epiglottis with arytenoid cartilage

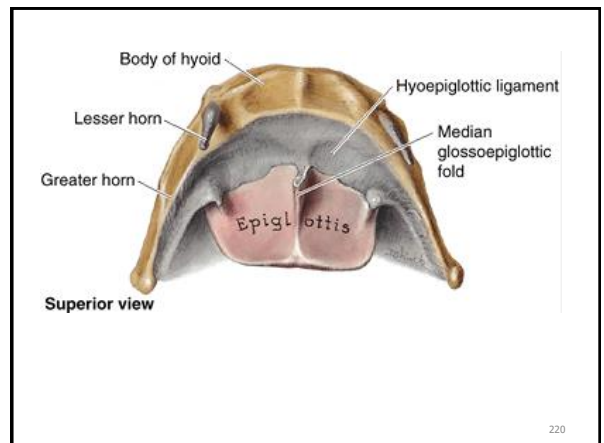
## ▪ Median & lateral glosso-epiglottic folds

- ✓ Connects the anterior surface of the epiglottis to the dorsum the tongue

## ▪ Crico-tracheal ligament

- ✓ Connects first ring of trachea to cricoid

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## Muscles of larynx

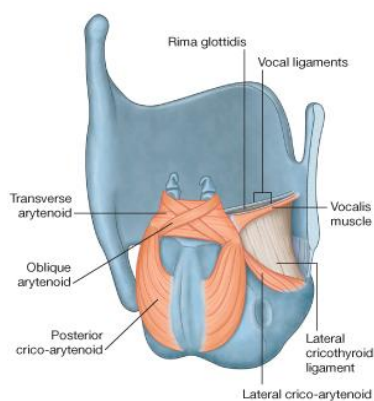
- **Extrinsic muscles** – move larynx during swallowing
  - Infrahyoid muscles – depress hyoid and larynx
  - Suprahyoid muscles – elevates hyoid and larynx
  - Thyrohyoid – depress hyoid and elevate thyroid cartilage
- **Intrinsic muscles** – move laryngeal parts
  - Alter length and tension of **vocal folds** and size and shape of rima glottidis
  - Divided based on their action on inlet of larynx

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## ▪ Muscles closing the inlet of the larynx

- ✓ **Transverse arytenoid muscle**
  - unpaired
  - attaches to the posterior aspects of the 2 arytenoid cartilage
  - Its contraction will adduct the arytenoid cartilages, vocal processes and vocal cords
- **Oblique arytenoid**
  - Superficial to transverse
  - 2 bundles cross each other, each arises from the muscular process of one arytenoid to the apex of the other arytenoid

222



223

## ▪ Abductor of vocal folds

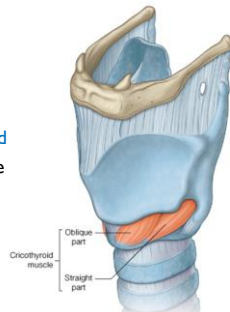
- ✓ **Posterior crico-arytenoid muscle**
  - From the posterior surface of the lamina of the cricoid cartilage to the posterior aspect of the muscular process of the arytenoid cartilage
  - Rotate arytenoids thus abduct the vocal processes and open the rima glottidis
- ✓ **Thyro-epiglottic muscle**
  - Connect the deep surface of lamina of thyroid cartilage with sides of the epiglottis
  - Superior fibers of the thyroarytenoid muscles pass into the aryepiglottic fold, and some of them reach the epiglottic cartilage; These fibers constitute the thyroepiglottic muscle
  - Pulls the epiglottis forward and widens the inlet

224

## ▪ Tensor of vocal cords

### • Cricothyroid muscle

- ✓ Arise from lower border of the **cricoid** and insert into the **inferior horn of thyroid**
- ✓ Its contraction tilts the thyroid cartilage anteriorly
- ✓ resulting in tension of the vocal cords

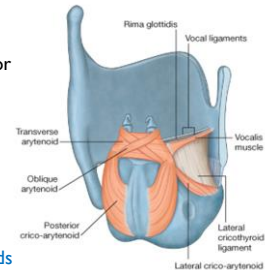


225

## ▪ Adductors of vocal folds

### ✓ Lateral crico-arytenoid muscle

- Attaches from the superior aspect of the arch of the cricoid to the **anterior surface of the muscular process of the arytenoid cartilage**
- Rotate the arytenoid cartilages medially and approximate the vocal folds

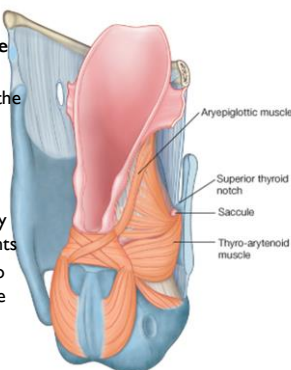


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## ▪ Relaxer of vocal folds

### ✓ Thyro-arytenoid muscle

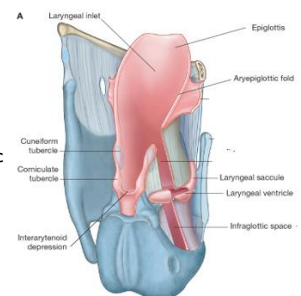
- Arise from the internal surface of the angle of the thyroid cartilage and insert into the **lateral aspect of the arytenoid cartilage**
- Pull arytenoid anteriorly thus relax vocal ligaments
- The portion adjacent to the vocal ligament is the vocalis muscle



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## Cavities of the larynx

- ✓ Extend from inlet of larynx to inferior border of cricoid
- ✓ **Superior laryngeal aperture (inlet)**; is obliquely oriented
  - Bounded by;**
    - **Laterally:** the aryepiglottic folds,
    - **Posterior:** interarytenoid fold and
    - **Anteriorly:** epiglottis

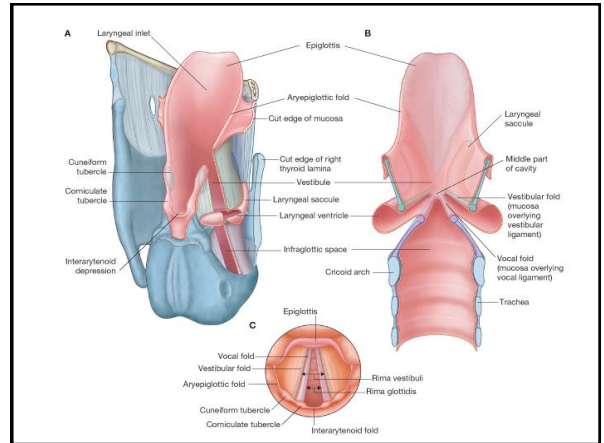


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## ■ Cavities of the larynx

- ✓ Two folds of mucous membrane – vestibular (upper) and vocal (lower) folds
  - **Rima vestibuli** – spaces between the vestibular folds (false vocal cords)
  - **Rima glottidis** – spaces between the vocal folds
- ✓ The two folds divide the cavity into 3 parts
  - Vestibule – superior to vestibular fold
  - Ventricle – between vestibular and vocal folds
  - Infraglottic cavity – below vocal folds

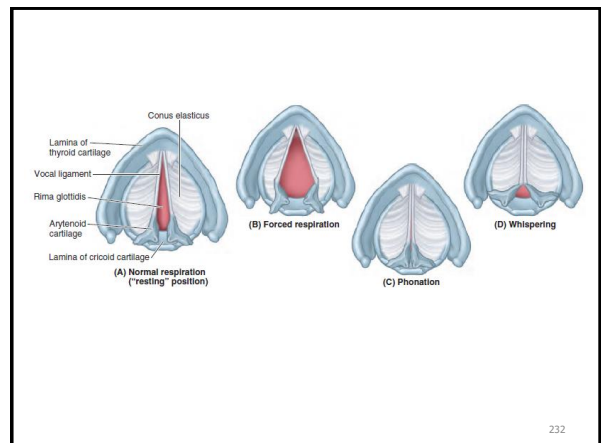
229



## ■ Vocal folds

- ✓ Concerned with production of sound
- ✓ Wedge-shaped; apex pointed medially and base lies against thyroid lamina
- ✓ Consists of vocal ligament, conus elasticus, muscle and mucous membrane
- ✓ Vocal folds, rima glottidis and narrow part of larynx together are called glottis
- ✓ Shape of rima glottidis vary according to position of vocal folds
  - Narrow during ordinary breathing
  - Wide during forced breathing
  - Linear slit during speaking

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- Voice pitch can be changed by varying tension and length of vocal folds, width of rima glottidis and intensity of expiratory force
  - Lower in males due to greater length of vocal folds
- **Vestibular folds** – consists of 2 folds of mucous membrane enclosing vestibular ligaments
  - Function - protective; close larynx during swallowing

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## Blood supply

### Arteries

- ✓ **Superior laryngeal artery** – branch of superior thyroid; runs with internal laryngeal nerve and pierce thyrohyoid membrane and supply interior of larynx **up to vocal folds**
- ✓ **Inferior laryngeal artery** – branch of inferior thyroid; supply mucosa and muscles **below vocal folds**

### Veins

- ✓ **Superior laryngeal vein** – drain into superior thyroid
- ✓ **Inferior laryngeal vein** – drain into inferior thyroid

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## Innervation

### ✓ Motor

- All intrinsic muscles are supplied by **recurrent laryngeal nerve** **except cricothyroid** (by **external laryngeal**)

### ✓ Sensory

- **Internal laryngeal** – above vocal folds
- **Recurrent laryngeal** – below vocal folds

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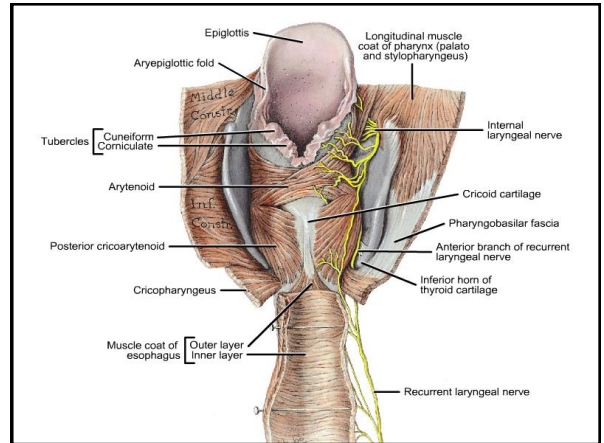
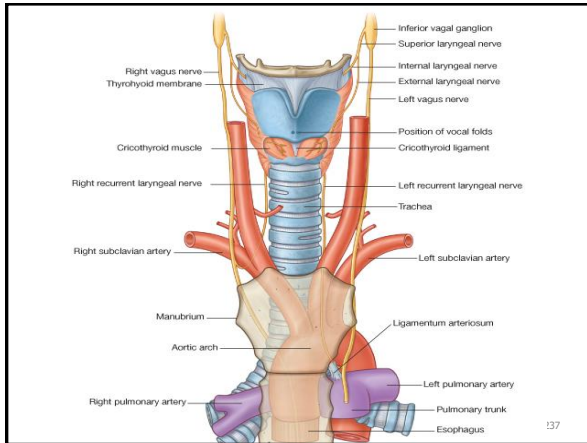
## Superior laryngeal nerve

- ✓ A branch of vagus, **divided into 2 terminal branches**
- ✓ **Internal laryngeal nerve** – pierces thyrohyoid membrane and supply laryngeal mucosa above vocal folds
- ✓ **External laryngeal nerve** – descends posterior to sternothyroid and supply inferior constrictor and cricothyroid

## Recurrent laryngeal nerve

- ✓ Ascends in a groove between trachea and esophagus and gives branches **to pharynx, esophagus and trachea**
- ✓ Supplies all intrinsic muscles of larynx except cricothyroid and mucous membrane below vocal folds
- ✓ Terminal part **enters larynx below inferior constrictor** and divide into anterior and posterior branches

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## ▪ Lymphatic drainage

- ✓ Superior to vocal folds – accompany superior laryngeal vessels and drain into superior deep cervical nodes
- ✓ Inferior to vocal folds – to inferior deep cervical nodes through prelaryngeal, pretracheal and paratracheal nodes

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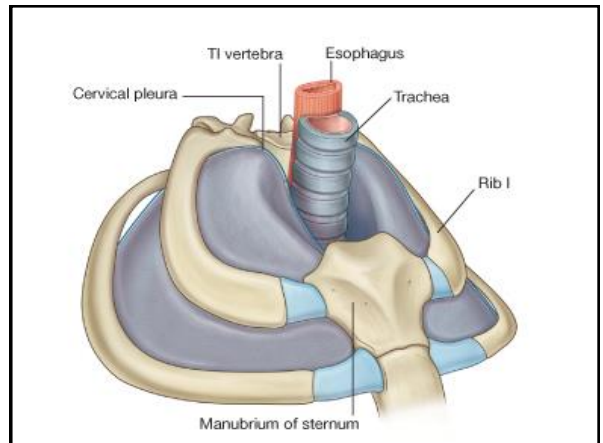
## Root of the neck (thoracocervical region)

240

## Introduction

- Junction area between thorax and neck
- Boundaries
  - Laterally - first rib & their costal cartilage
  - Anteriorly - manubrium of sternum
  - Posteriorly - body of T1

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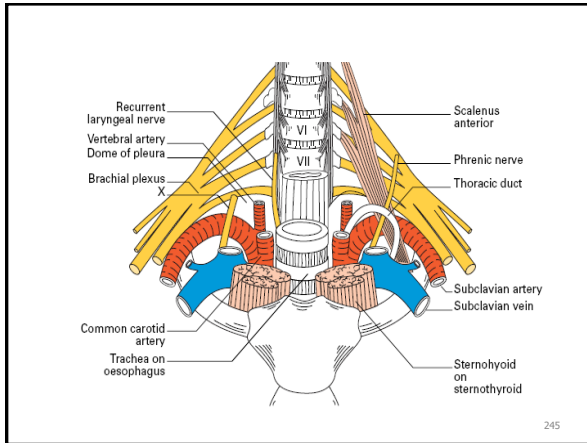
## Arteries in the root of neck

- Brachiocephalic trunk
  - on right side
  - behind sternoclavicular joint divide into common carotid artery and subclavian artery
  - has no branch but a variable artery called thyroidea ima artery may be found

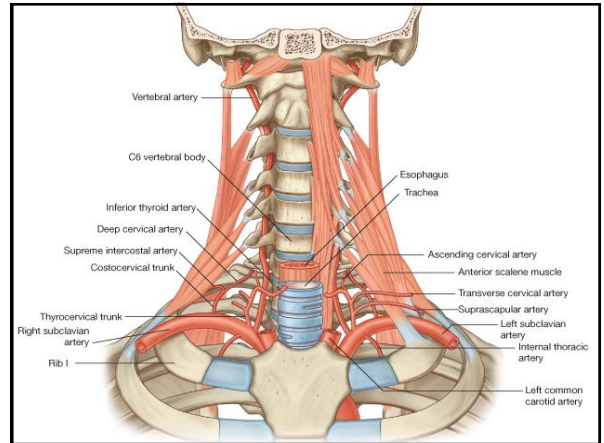
243

- Subclavian artery
  - Arise from brachiocephalic artery on right side and from arch of aorta on left side
  - runs over the apex of the lung, posterior to the scalene anterior and superior to first rib
  - it can be divided into 3 parts:
    - Medial to scalenus anterior
    - Posterior to scalenus anterior
    - Lateral to scalenus anterior
  - Branches: vertebral, internal thoracic, thyrocervical trunk, costocervical trunk and dorsal scapular

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### • Vertebral artery

- The largest branch
- Supplies brain, spinal cord, meninges, muscle and bones
- Four parts
- First part
  - From origin to C6
  - Course upwards and backwards between scalenus anterior and longus colli
- Second part
  - Passes upwards through transverse foramina of C1-C6

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### • Third part

- In suboccipital triangle
- Winds medially and enters skull through foramen magnum

### • Fourth part

- Ascends medially to reach the front of medulla
- At lower border of pons unites with the other artery to form basilar artery

### • Branches

- Cervical branches to spinal cord, meninges and vertebrae
- Muscular branches to suboccipital triangle
- Cranial branches to brain

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- **Thyrocerivcal trunk**

- Divide into inferior thyroid, suprascapular and transverse cervical arteries
- Inferior thyroid artery
  - branches
    - glandular - supplies the inferior pole of thyroid gland; ascending branch anastomoses with posterior branch of superior thyroid artery to supply parathyroid gland
    - ascending cervical artery - supplies the scalenus anterior and the longus colli muscles
    - inferior laryngeal artery – to larynx
    - other branches supply surrounding structures
- Transverse cervical artery and suprascapular arteries
  - pass anterior to scalenus anterior as they run into the posterior triangle of the neck

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- **Internal thoracic artery**

- runs inferiorly into thorax parallel to sternum and gives off anterior intercostal arteries

- **Costocervical trunk**

- passes posteriorly over the apex of the lung and the neck of first rib and divide into 2 branches
  - superior intercostals – gives first and second posterior intercostals arteries
  - Deep cervical artery – ascends between semispinalis capitis and cervicis and anastomoses with occipital and vertebral arteries

- **Dorsal scapular artery**

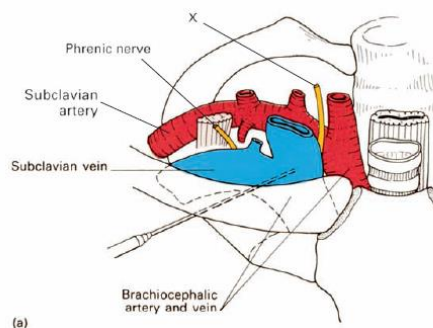
- Pass laterally and backwards to scapula

250

- **Applied anatomy**

- Third part of subclavian artery can be compressed against first rib to control bleeding in upper limb
- Pressure is applied downwards, backwards and medially in angle between SCM and clavicle

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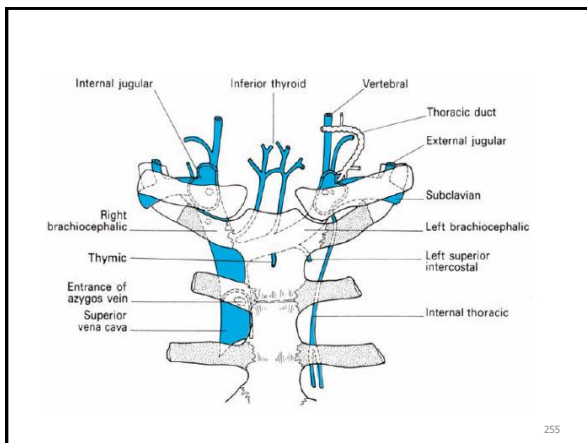
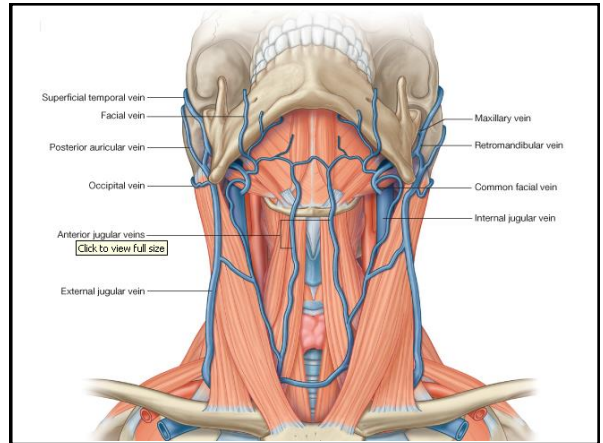


252

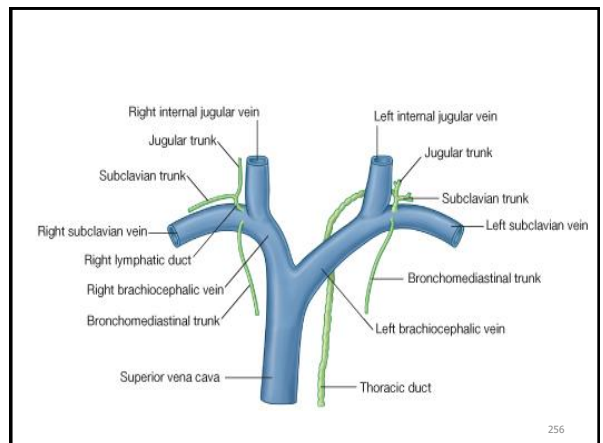
## Veins in the root of neck

- Anterior jugular vein
  - Arise in submental region from submental venous plexus
  - Descends between anterior medial line and anterior border of SCM
  - At root of neck turns laterally and opens into external jugular vein
  - Superior to sternum right and left veins are connected by jugular venous arch
- Subclavian vein
  - Continuation of axillary vein from lateral border of first rib
  - Joins internal jugular vein at medial border of scalenus anterior to form brachiocephalic vein
  - Tributaries – external jugular vein, dorsal scapular vein, thoracic duct and right lymphatic duct

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## Nerves in the root of the neck

### • Vagus nerve

#### – Course and relation

- Leaves cranial cavity through jugular foramen where it is joined by cranial root of accessory nerve
- descends with in carotid sheath
- At root of neck the right cross subclavian and turn medially to trachea whereas the left one passes between common carotid artery and subclavian artery

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### • Branches

- Auricular branch – supply root of auricle and posterior half of external auditory canal
- Pharyngeal branch – contains chiefly fibers of cranial root of accessory nerve and forms pharyngeal plexus which supply muscles of pharynx and palate
- Superior laryngeal nerve
  - Divide into external and internal laryngeal nerves
  - Internal laryngeal branch - pierces the thyrohyoid membrane and enter larynx to supply laryngeal mucosa above vocal cords and is involved in the cough reflex
  - External laryngeal branch - runs on the lateral aspect of the larynx, accompanies superior thyroid artery, to innervate the cricothyroid muscle which tenses vocal cords during vocalization

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#### – Right recurrent laryngeal nerve

- arises from the vagus in front of subclavian artery and loops under the artery to ascend to the larynx between the trachea and the esophagus

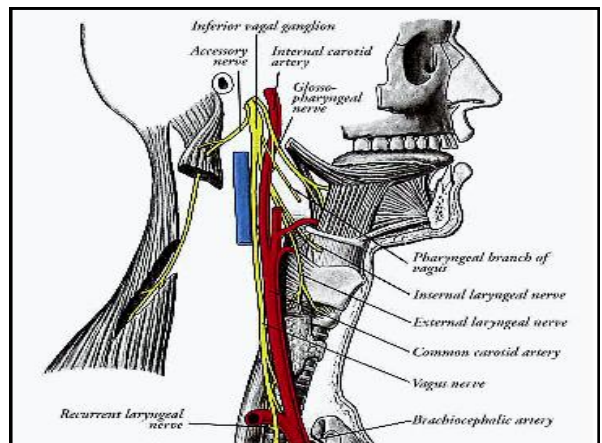
#### – Left recurrent laryngeal nerve

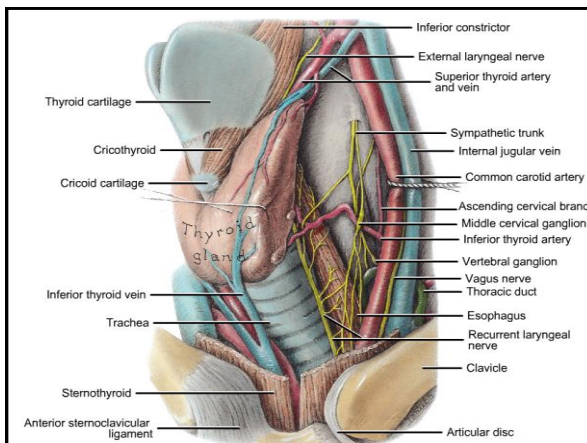
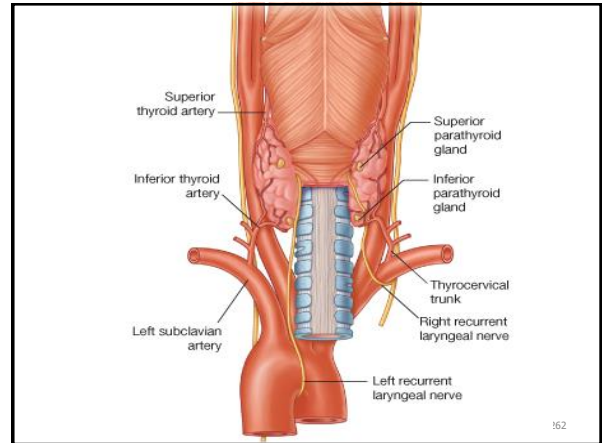
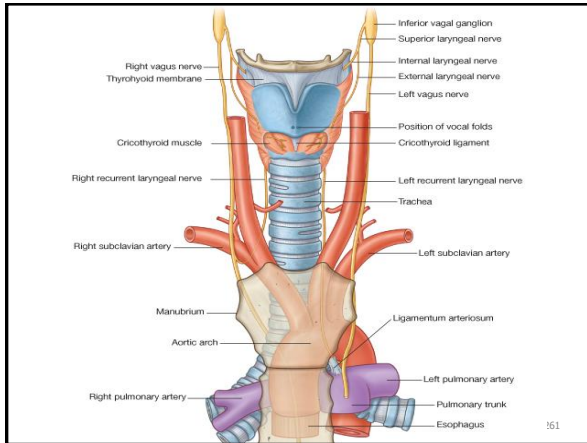
- arise at arch of aorta
- loops around ligamentum arteriosum and reach tracheoesophageal groove

#### – They supply

- all intrinsic muscles of larynx except cricothyroid
- mucosa of larynx below vocal cords
- cardiac branch to deep plexus
- branches to trachea, esophagus and to inferior constrictor

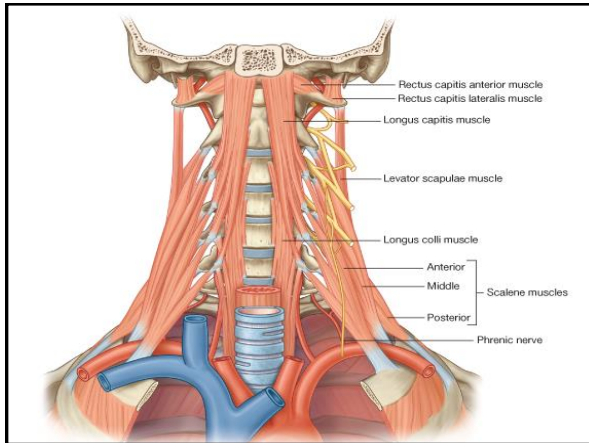
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## Phrenic nerve

- Supplies diaphragm, pleura and pericardium
- Arise chiefly from C4 with contribution from C3 and C5
- Formed at lateral border of scalenus anterior
- Runs downwards over scalenus anterior, on cervical pleura and crosses internal thoracic artery medially to enter the thorax behind first costal cartilage
- Left phrenic nerve crosses in front of subclavian artery



## Sympathetic trunk

- Longitudinal strands of autonomic nerve fibers associated with sympathetic ganglia
- Located anterolateral to the vertebral column, in front of prevertebral muscles
- Contains three pairs of cervical ganglia (superior, middle & inferior)
- The ganglia receive preganglionic fibers from superior 4 thoracic spinal nerves through white rami communicantes
- Postganglionic fibers pass to cervical structure through cervical spinal nerves or as direct branch to viscera
- Branches to the head run with arteries especially with internal & external carotid arteries

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## Inferior cervical ganglion

- Collection of neuronal cell bodies at the level of superior border of the neck of the first rib, transverse process of C7
- Mostly fused with first thoracic ganglion to form large ganglion called cervicothoracic (stellate) ganglion
- Branches
  - Rami communicantes to ventral rami of C7 & C8 nerves
  - Vertebral branch – form plexus around vertebral artery
  - Subclavian branch – form plexus around subclavian artery
  - Inferior cardiac branch – to deep cardiac plexus

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## Middle cervical ganglion

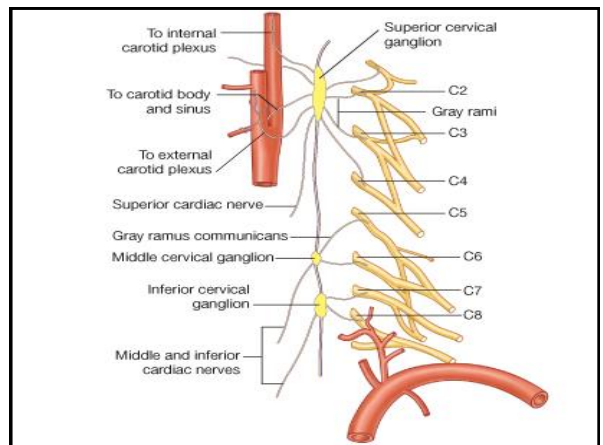
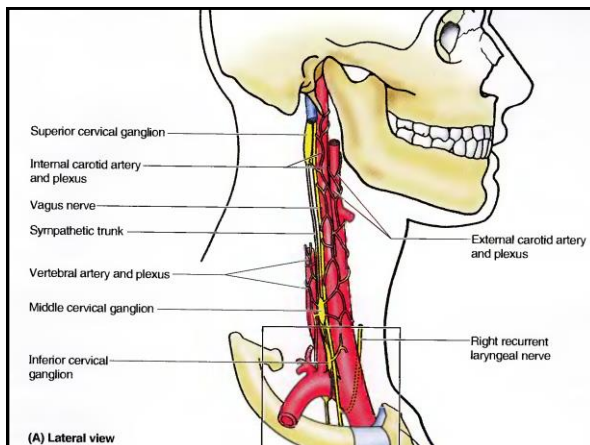
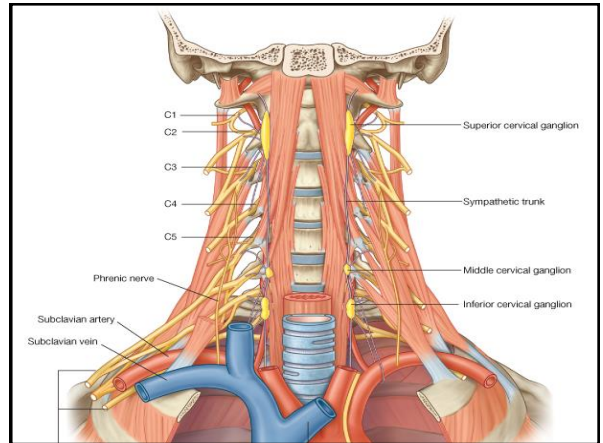
- At the level of transverse process of C6 on the anterior aspect of inferior thyroid artery
- Branches
  - To inferior cervical ganglion directly or through a loop around subclavian artery (ansa subclavia)
  - Rami communicantes to C5&6
  - Thyroid branch to thyroid and parathyroid
  - Tracheal and esophageal branch
  - Middle cervical cardiac branch

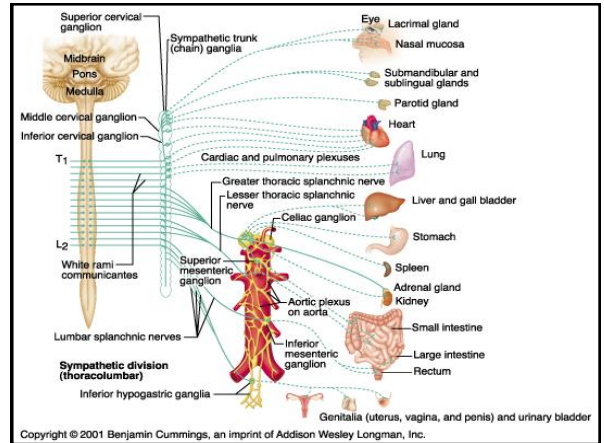
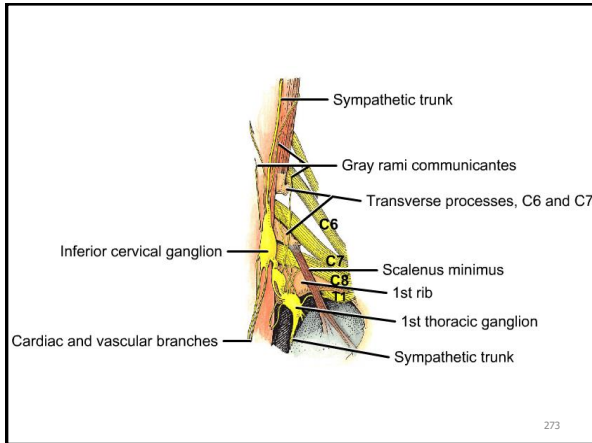
268

## Superior cervical ganglion

- The largest, 2-3 cm long, at the level of atlas and axis
- Communicate with 9th, 10th and 12th cranial nerves
- Branches
  - Rami comm to upper 4 cervical nerves
  - Internal carotid plexus
  - External carotid plexus
  - Pharyngeal branch to pharyngeal plexus
  - Cardiac branch

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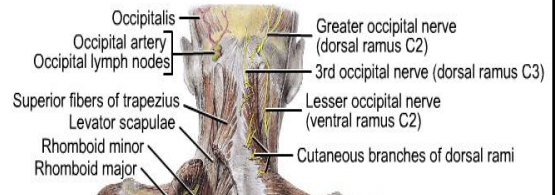
## Back of the neck

- Nerves
  - Cutaneous nerves – medial branches of dorsal rami of C2-5
    - Greater occipital – medial branch of dorsal ramus of C2
    - Lesser occipital – medial branch of ventral ramus of C2
    - Third occipital – medial branch of dorsal ramus of C3
- Ligamentum nuchae
  - strong triangular fibroelastic septum
  - Attachments
    - Base – to the external occipital protuberance
    - Apex – to spine of C7
    - Posterior – to spines of cervical vertebrae
    - Anterior – to investing deep fascia of the neck
  - Origin for – trapezius, splenius & rhomboid minor

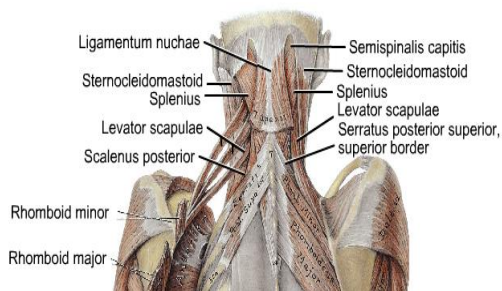
- Muscles of the back – 4 layers

- 1st layer – trapezius
- 2nd layer – splenius & levator scapulae
- 3rd layer – semispinalis capitis & longissimus capitis
- 4th layer – sub occipital muscles, multifidus, rotators & interspinalis

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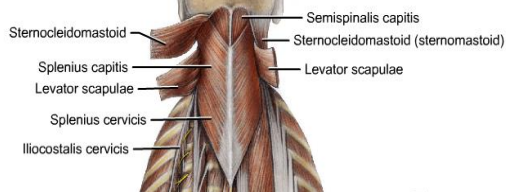


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- **Splenius**

- Origin – lower half of ligamentum nuchae, spines of C7 & T1-6
- Insertion – two parts
  - Upper part – splenius capitis - lateral part of mastoid process & superior nuchal line
  - Lower parts - splenius cervicis – upper part of transverse process of C2-3
- Nerve supply – dorsal rami of C4-8
- Action
  - one muscle – bend the head and neck to the same side & turns the face to the same side
  - Two muscles - extend the head and neck

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### • Semispinalis

– Origin - from the transverse processes of T4-6

– Insertion – three parts

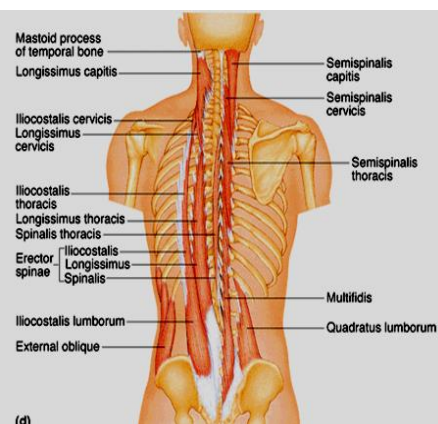
- Semispinalis capitis - deep to the splenius capitis into the area between superior and inferior nuchal line
- Semispinalis cervicis – deep to the semispinalis capitis into spines of C2-4
- Semispinalis thoracis - upper thoracic spines

– Nerve supply – dorsal rami of cervical nerves

– Action

- one muscle – bend the head to the same side & turns the face to the opposite side
- Two muscles - extend the neck

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### • Longissimus capitis

– lateral to the semispinalis & deep to the splenius capitis

– Origin - from the transverse processes of T4-6

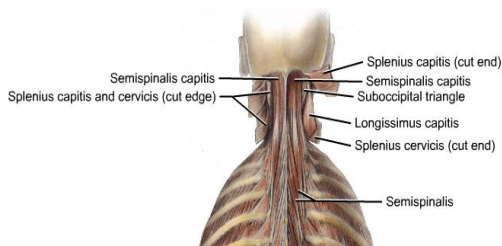
– Insertion – mastoid process

– Nerve supply – dorsal rami of cervical nerves

– Action

- one muscle – bend the head to the same side & turns the face to the opposite side
- Two muscles- extend the neck

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- Suboccipital muscles
  - four muscles on each side
  - Rectus capitis posterior major
    - Origin - spine of axis
    - Insertion – lateral area below the inferior nuchal line
  - Rectus capitis posterior minor
    - Origin – spinous tubercle of atlas
    - Insertion – medial part of the skull below the nuchal line

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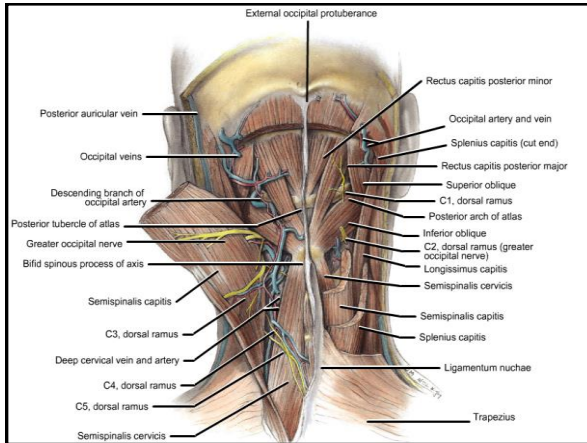
- Inferior oblique
  - Origin - spine of axis
  - Insertion – lateral area below the inferior nuchal line
- Superior oblique
  - Origin – transverse process of atlas
  - Insertion – lateral area of the skull between the superior & inferior nuchal lines
- Action of suboccipital muscles – act as ligament to stabilize the head, atlas & axis

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## Suboccipital triangle of the neck

- Boundaries
  - Above and medially- rectus capitis posterior major and minor
  - Laterally – superior oblique
  - Inferiorly – inferior oblique
  - Roof
    - Medially – semispinalis
    - Laterally – splenius capitis
  - Floor - posterior arch of atlas and posterior atlantooccipital membrane

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### • Contents

- 3rd part of vertebral artery
- suboccipital nerve (dorsal ramus of C1)
- suboccipital plexus of veins

### • 3rd part of vertebral artery

- Begins at the transverse foramina of the atlas & pass laterally back ward which directs medially to enter the posterior cranial fossa
- Lies in the groove of posterior arch of atlas
- Supplies muscles in the back

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### • Suboccipital nerve

- Enter the triangle between posterior arch of atlas & 3rd part of vertebral artery
- Branches
  - Muscular – 4 occipital and semispinalis muscle
  - Communicating – to greater occipital
  - Cutaneous – some times

### • Suboccipital plexus of veins

- Collect blood from occipital veins, internal vertebral venous plexus and emissary veins
- Drain into deep cervical vein & vertebral venous plexus

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### • Occipital artery

- Origin from external carotid
- Parallel to the posterior belly of digastric muscle up to the mastoid process, passes superficial to internal carotid, internal jugular, vagus, accessory & hypoglossal
- After the mastoid process, crosses the apex of posterior triangle of the neck
- Branches – muscular, auricular, mastoid & descending

### • Deep cervical artery

- Origin from the costocervical trunk
- Anastomoses with descending branch of occipital artery

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- **Posterior rami of cervical nerves**

- Each with medial and lateral branches
- Supply
  - muscles of the back of the neck & intervertebral disc
  - skin of the back of the neck
- **Greater occipital**
  - medial branch of posterior ramus of C2
  - supply skin of the back of the neck and motor for semispinalis
- **Third occipital**
  - medial branch of posterior ramus of C3
  - supply skin of the back of the neck up to the vertex of the scalp

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## Deep structures in neck

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## Prevertebral muscles

- Extend from skull base to superior mediastinum, covering vertebral column
- Covered by prevertebral fascia
- Are flexors of head and neck
- Supplied by ventral rami of cervical nerves

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- **Longus colli (cervicis)**

- Upper and lower part is oblique and middle part is vertical
- Origin
  - upper oblique part – transverse processes of C3-C5
  - lower oblique part - bodies of upper 2 or 3 thoracic vertebrae
  - middle vertical part - bodies of C5 – T3
- insertion
  - upper – anterior tubercle of atlas
  - lower – anterior tubercles of C5 and C6
  - middle – bodies of C2-C4
- nerve supply – C3 – C8

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- **Longus Capitis**

- Broad superiorly and narrow inferiorly
- Origin – transverse processes of C3- C6 vertebrae
- Insertion – base of skull
- Nerve supply – C1 – C3

- **Rectus Capitis Anterior**

- Origin – lateral mass of atlas
- Insertion – base of skull
- Nerve – C1

- **Rectus Capitis Lateralis**

- Origin – transverse process of atlas
- Insertion – occipital bone
- Nerve supply – C1 and C2

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